



TE 01_STLA M6 Connectivity South Bound Interface Specification

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Company Confidential

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1. Revision History

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0.1	27-Nov-2025	All	Initial template release
0.2	28-Nov-2025	All	Addressed peer review comments
0.3	28-Nov-2025	Sec 8.2.1	Addressed SQA review comments.
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1.1	12-Dec-2025	Sec 17,16.3.1	Addressed the comments from STLA and Release to STLA
3.0	19-Dec-25	Sec 13.11 to 13.13	Trip Payload unnecessary parameters are removed. (3.0 - Aligning to proto version)

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3. ACRONYMS AND ABBREVIATIONS

Acronym	Abbreviations
CAN	Controller Area Network
CVIP	Connected Vehicle India Platform
DLC	Data Length Code
DUP	Duplicate
eSIM	Embedded SIM
KYC	Know Your Customer
ECU	Electronic Controller Unit
EID	Embedded Identity Document
EOL	End of Line
FOTA	Firmware Over the Air
GPS	Global Positioning System
HTTPS	Hypertext Transfer Protocol Secure
ICE	Internal Combustion Engine
IGN	Ignition
IMEI	International Mobile Equipment Identity
JSON	JavaScript Object Notation
MQTT	Message Queuing Telemetry Transport
MSISDN	Mobile Station International Subscriber Directory Number
NTP	Network Time Protocol
PDU	Protocol Data Unit
QoS	Quality of Service
DONGLE	Telemetry Box
TLS	Transport Layer Security
VIN	Vehicle Identification Number

4. Glossary

Term	Definition
Message ID	Message IDs are UUIDS which are generated based on the current timestamp. CVIP is using JAVA package UUIDS and function UUIDS.timeBased() for generating the message ID.
Correlation ID	Correlation ID in a particular response is the message id of the corresponding command received.
Communication Frequency	The frequency at which message is transmitted to cloud
Sampling Frequency	The frequency at which Dongle collates/receives the CAN frames.
Partial Sleep	IGN off state is considered as partial sleep

5. Scope of the Document

This document captures the following aspects for Connected Vehicle India Platform (CVIP):

- Connectivity handshake between the DONGLE and CVIP
- Dongle Activation process and state transitions
- MQTT client and server configurations
- Telemetry Streams definitions – Vehicle & Location
- Alerts & Events definitions – Vehicle Alerts, Events, diagnostic alerts, Emergency & safety alerts

6. DONGLE CVIP Connectivity

6.1 Connectivity protocols between DONGLE and CVIP

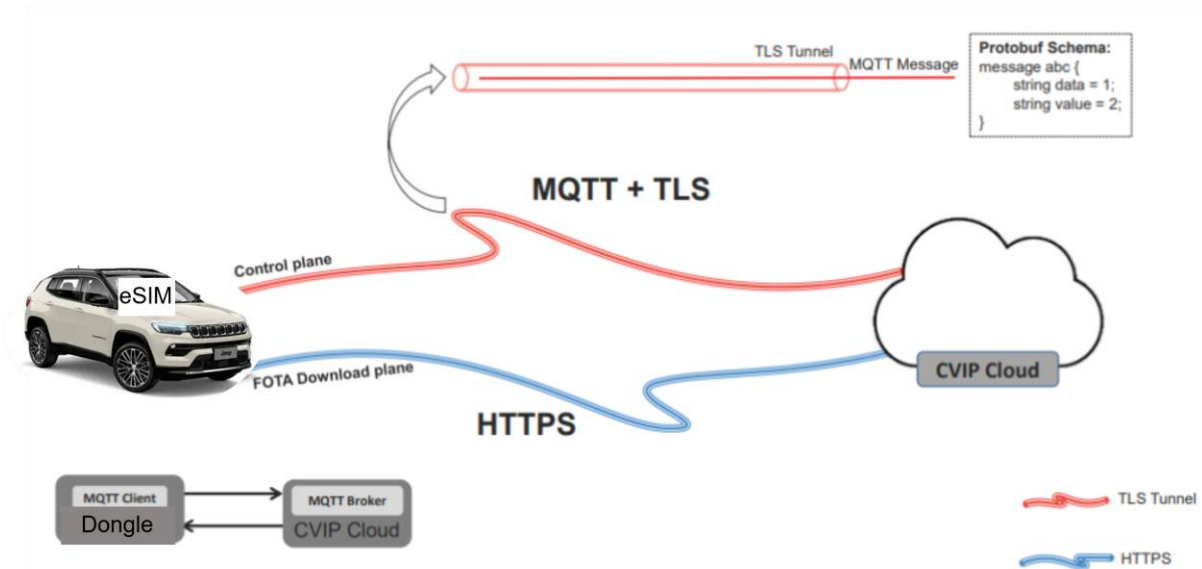


Figure 1: Connectivity between Dongle and CVIP Control plane data uses MQTT + TLS
FOTA Download uses HTTPS

Note: for MQTT+TLS and HTTPS, self-signed certificate shall be used during development

6.2 MQTT

MQTT shall be the protocol interface between the CVIP and Dongle. The scalable and enterprise grade MQTT broker (VerneMQ) used on the CVIP currently supports only MQTT v3.1.1.

QoS level-1 shall be used for the MQTT publish/subscribe requests for all topics. Use of QoS level-2 will not be evaluated as it involves overheads and needs to be prototypes and performance benchmarked.

6.2.1 MQTT Connection Request

The following table lists the flags used in the MQTT connection request:

Field	No. of bits	Default Values used	Description
Client Id	17	VIN/IMEI	Unique Client Id. <i>NOTE : Dongle is expected to use IMEI as client ID for</i>

Field	No. of bits	Default Values used	Description
			<i>MQTT connection till VIN is successfully written in the Dongle. After successful VIN write operation Dongle is expected to use VIN as the client ID for MQTT connection.</i>
Will Retain	1	0	Will Flag set to 0.
Will QoS Level	2	1	Will Flag set to 0.
Will Flag	1	0	The Will message MUST NOT be published when the Network Connection ends.
Clean Session	1	0	The Client and Server MUST NOT discard any previous Session and start a new one. The Session continues after recovering from the Network failure.

Client Id shall be set to VIN (configurable in the DONGLE) while establishing connection with the MQTT broker.

All MQTT related parameters shall be made configurable and can be set to default initially.

6.2.2 MQTT Publish Request

The following table lists some of the critical flags used in the MQTT publish request:

Field	No. of bits	Value	Description
DUP Flag	1	0/1	First transmission (0) or Re-transmission (1)
QOS level	2	1	QoS – 1; At least one delivery (Acknowledged delivery)
Retain	1	0	Server will not store the Application Message and its QoS.

The publish request payload shall include the MQTT topic and a message identifier along with the actual message.

6.2.3 MQTT Subscribe Request

The following table lists some of the critical the fields used in the subscribe request MQTT message:

Field	No. of bits	Value	Description
Topic	Variable(n)		The name of the topic being subscribed to. I.e Refer to sections 8.1 and 8.2

Field	No. of bits	Value	Description
Requested QoS	8	0x1	QoS – 1; Atleast once delivery (Acknowledged delivery)

6.2.4 MQTT Keep Alive

The clients connecting to MQTT broker in CVIP are expected to configure the keep alive time of 10 minutes. This means that the client is expected to publish at least one message within 10 minutes from the last communicated message. If there is no communication of message for the past 10 mins, then the client is expected to publish a PingReq message.

The MQTT broker shall wait for 15 minutes (1.5 times the keep alive timer) to abort a MQTT session if there is no communication from the client.

6.2.5 MQTT LWT configuration

Dongle is expected to configure the deviceJoined topic as LWT topic in the MQTT client configuration.

Topic Name - /dongle/<VIN>/MQTTPROTOBUF/deviceJoined

LWT message shall be configured as the deviceJoined message with DongleOperatingState configured as “DISCONNECTED”.

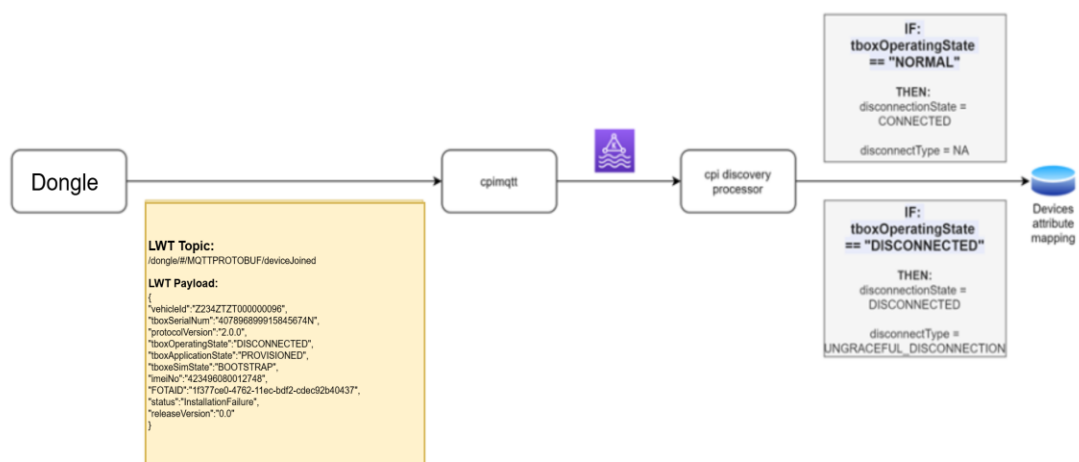


Figure 2: MQTT LWT flow

6.2.6 TLS Version and Curve Configuration

Preprod Environment

Version – 1.2

Enabled Curves - secp256k1:prime256v1:secp384r1

Prod Environment

Version 1.2 and 1.3

Enabled Curves - secp256k1:prime256v1:secp384r1

7. Dongle Activation

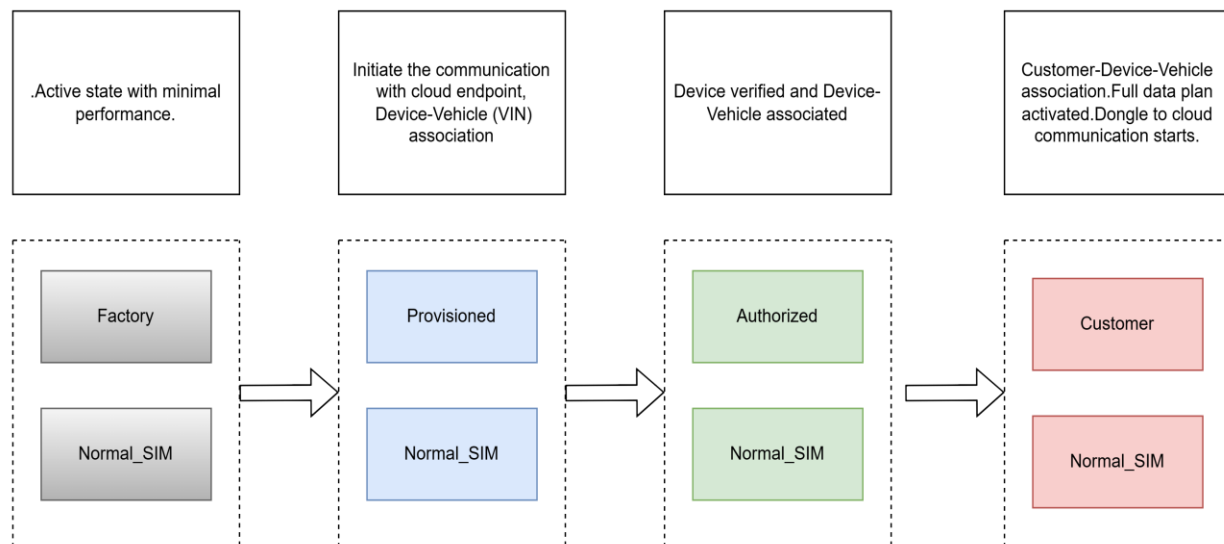


Figure 3: DONGLE states

7.1 Dongle State Update

- Triggers will be sent from CVIP to Dongle for state transitions to PROVISIONED, AUTHORIZED, and CUSTOMER. Message formats are detailed in section Dongle State Update.
- Once updated by CVIP, Dongle shall never move to a previous state.

7.2 Public Key Infrastructure

7.2.1 IMEI Upload

SB API Interface ID : SB-DONGLE_ACTIVATION-001

- In order to allow ping verification for whitelisted IMEI, Dongle Manufacturer shall communicate the IMEIs which is planned for ping verification.
- Rest API shall be exposed to Dongle Manufacturer which shall be used by their factory team to communicate the IMEIs.
- IMEIs input API shall be authenticated using the time-limited access tokens. Access tokens shall be generated using username and password.

- The Rest API shall allow communication from the Dongle Manufacturer factory server IP address. The API shall not allow access from any other IP address.
- IMEI is 15-digit numeric value. First eight digits of the IMEI is the TAC value which is uniquely allocated for Dongle Manufacturer. The remaining 7 digits is a rolling counter value.

7.2.1.1 Assumptions

- First eight digits of the IMEI is the TAC value which is uniquely allocated for Dongle Manufacturer. The remaining 7 digits is a rolling counter value.
- The TAC value which is first eight digits of the IMEI is assumed as “35674100” for existing dongles.
- The TAC value for newly onboarding dongle is expected to be provided by the Dongle Manufacturer.

NOTE_: Assumptions in section 8.2.1.1 are to be determined after discussion with Dongle manufacturer and then updated in the next version of the document.

7.2.2 Common Ping Verification Certificate Generation

SB API Interface ID : SB-DONGLE_ACTIVATION-002

- Device manufacturer shall use the certificate creation API to generate the ping verification certificate.
- Ping verification certificate generation is allowed only once by the CVIP server.
- Ping verification by Dongle shall be authenticated using ping verification certificate which is common for all devices. Hence, platform shall allow ping verification only with ping verification certificate
- Ping verification shall be successful with CVIP if the common name in the ping verification certificate is “CVIP-Jeep-CommonCertificate-Dongle”.
- Server shall validate if the public key received in the CSR is a valid ECC-256 bit key
- The CSR shall include the following mandatory parameters.

CA Subject DN	Optional/ Mandatory	Remarks
Common Name	M	CVIP-Jeep-CommonCertificate-Dongle (Max Char Length : 64)
Organisation (O)	M	Legal Name of the CA (Max Char Length : 64)
Country (C)	M	Country code as per the verified CAs head office or registered office address (Max Char Length : 2 (“EG”))

7.2.3 Supplier Feed Upload

SB API Interface ID : SB-DONGLE_ACTIVATION-003

- Dongle Manufacturer Factory server shall communicate the list of IMEI's to CVIP.
- Supplier feed shall be a csv file which has the following fields.

Data	type	size	example	remark
Dongle_SN	string	18		Primary key
IMEI	string	15		
MSISDN	string	12		
HW_part_number	string	8		
Plant_Manufactured_Date	Long		1645703145	EPOC time
ICCID	string	20		
eUICCID	string	32		Optional Field
MCU_SW_version	string	9	MD0.01.00	
NAD_SW_version	string	9	ND0.01.00	
countryCode	Integer	4		
regionCode	Integer	4		
plantManufacturingCountryCode	Integer	4		Country code value to be fetched from “Look up table for Country Code with RegionCode Mapping”

Look up table for Region Code

regionCode – Value to be sent in Supplier Feed	Region Name
0	Default/Invalid
1	Middle East
2	Europe
3	Africa
4	Asia Pacific A
5	Asia Pacific B
6	Latin America A
7	Latin America B
8	North America
9	China
10	Russia

Look up table for Country Code with RegionCode Mapping

RegionCode	CountryCode - – Value to be sent in Supplier Feed	CountryName
0	0	Default/Invalid
4	1	India
4	2	Nepal
4	3	Bhutan
4	4	Bangladesh
4	5	Myanmar
4	6	Laos
4	7	Thailand
4	8	Cambodia
4	9	Vietnam
4	10	Indonesia
5	11	Malaysia
5	12	Singapore
5	13	Philippines
5	14	Taiwan
5	15	Korea
5	16	Japan
9	17	China
6	18	Venezuela
6	19	Colombia
6	20	Ecuador
6	21	Peru
6	22	Bolivia
6	23	Chile
6	24	Paraguay
6	25	Argentina
6	26	Guyana
7	27	Suriname
7	28	Guiana
7	29	Brazil
7	30	Uruguay
4	31	Russia

- The supplier feed file shall be communicated as a multipart file through the supplier feed API.
- Supplier feed API shall be authenticated using the time-limited access tokens. Access tokens shall be generated using username and password.

- Supplier feed API shall allow communication from the Jeep Dealer server IP address. The API shall not allow access from any other IP address.
- Upon receiving the csv file, the server shall verify the IMEI which has passed the validation and map the device details like Dongle SN, ICCID etc. in the Device table.
- If the csv file has IMEI which is not part of the IMEIs input API, then such IMEI's shall also be communicated as an error response to Supplier feed API.
- If there are no errors in the csv input file, then the success response shall be communicated as a response to supplier feed API.

7.2.4 Token Generation for Certificate Signing request API

SB API Interface ID : SB-DONGLE_ACTIVATION-004

- After successful supplier feed communication to CVIP server, Dongle Manufacturer dispatches the Dongle to VPC/Dealer for vehicle fitment. Post the vehicle fitment, the Dongle is configured with the Vehicle identification number (VIN) and Certificate signing request to CVIP server is initiated..
- Dongle shall use the ping verification certificate to establish the MQTT + TLS session with CVIP server.
- Dongle shall request the access token using the topic and message name as mentioned below.

S. no	Topic	Publisher	Subscriber	Description
1	/dongle/{IMEI}/{protocol}{dataformat}/csrAccessTokenReq	Dongle	CVIP	Device will publish the request for CSR API access token
2	/dongle/{IMEI}/{protocol}{dataformat}/csrAccessTokenRsp	CVIP	Dongle	CVIP will respond with the Access token for CSR.

7.2.4.1 Sample Message from Dongle to CVIP for Access Token Request

dongle/356741000000021/MQTTPROTOBUF/csrAccessTokenReq

Message:

```
{
  "imei": "356741000000021"
}
```

7.2.4.2 Sample Message from CVIP to Dongle for Access Token Response

Topic Name:

/dongle/356741000000021/MQTTPROTOBUF/csrAccessTokenRsp

Message:

```
{
  "csrAccessTokenAck": "Success",

  "accessToken": "d7f913e3-74c5-30bd-ae2f-f8eac83d97bf"
}
```

7.2.5 Certificate creation from Dongle

SB API Interface ID : SB-DONGLE_ACTIVATION-005

- Once the token is successfully received by Dongle, it shall use the certificate creation API for generating the device specific certificate. The API shall have the access token as part of authorization header.
- The common name in the API body shall be a “IMEI-DongleSN”.
For eg. 356741000000321-A234ZTZT69322
- The server shall verify if the IMEI received in CSR has passed the following.
 - IMEI is part of the supplier feed data uploaded by Dongle Manufacturer.
 - Server shall also validate if the public key received in the CSR is a valid ECC-256 bit key
 - The CSR shall include the following mandatory parameters for first certificate creation.

CA Subject DN	Optional/ Mandatory	Remarks
Common Name	M	IMEI-DongleSN (Max Char Length : 64)
Organisation (O)	M	Legal Name of the CA (Max Char Length : 64)
Country (C)	M	Country code as per the verified CAs head office or registered office address (Max Char Length : 2 (“EG”))

- Server shall forward the CSR to EMCA using the create Certificate API after validating the request. The created certificate shall be responded to Dongle over the same API.
- Generated certificate shall be valid for 20 years.

- Once the Dongle receives the certificate, Dongle uses the certificate to establish the TLS session and communicate the Device joined message.

7.2.6 Vehicle Discovery

- When Dongle is fitted in the vehicle and VIN is provisioned, then Dongle state changes from factory to provisioned state.
- When turned on, Dongle sends Vehicle discovery with payload as detailed in section [Message format for topic “/dongle/{vinnum}/{protocol}{data format}/deviceJoined”](#) from Dongle to CVIP.
- CVIP, on successful validation of the payload communicates the device joined response and dongle state change from PROVISIONED to AUTHORIZED is triggered to dongle.
- Upon receiving the successful dongle state update command response CVIP updates the respective dongle state to AUTHORIZED.

7.2.7 Certificate Revocation

SB API Interface ID : SB-DONGLE_ACTIVATION-006

Certificate revocation list for Dongle can be accessed through the below URL.

CRL URL for Preprod - Platform server validation by Dongle

<https://mpki-int.emudhra.com/emCA/Repository/CVIPCloudPreprodICA.crl>

CRL URL for Prod - Platform server validation by Dongle

<http://cvip-psa-prod-cloud-sub-ca2-crl.s3.ap-south-1.amazonaws.com/crl/ee7c4dbd-fd81-4265-857f-57ddc38d76aa.crl>

7.3 Vehicle Discovery

SB API Interface ID : SB-DONGLE_ACTIVATION-007

Vehicle discovery happens at Jeep Dealer/VPC EOL.

7.3.1 Sequence Diagram

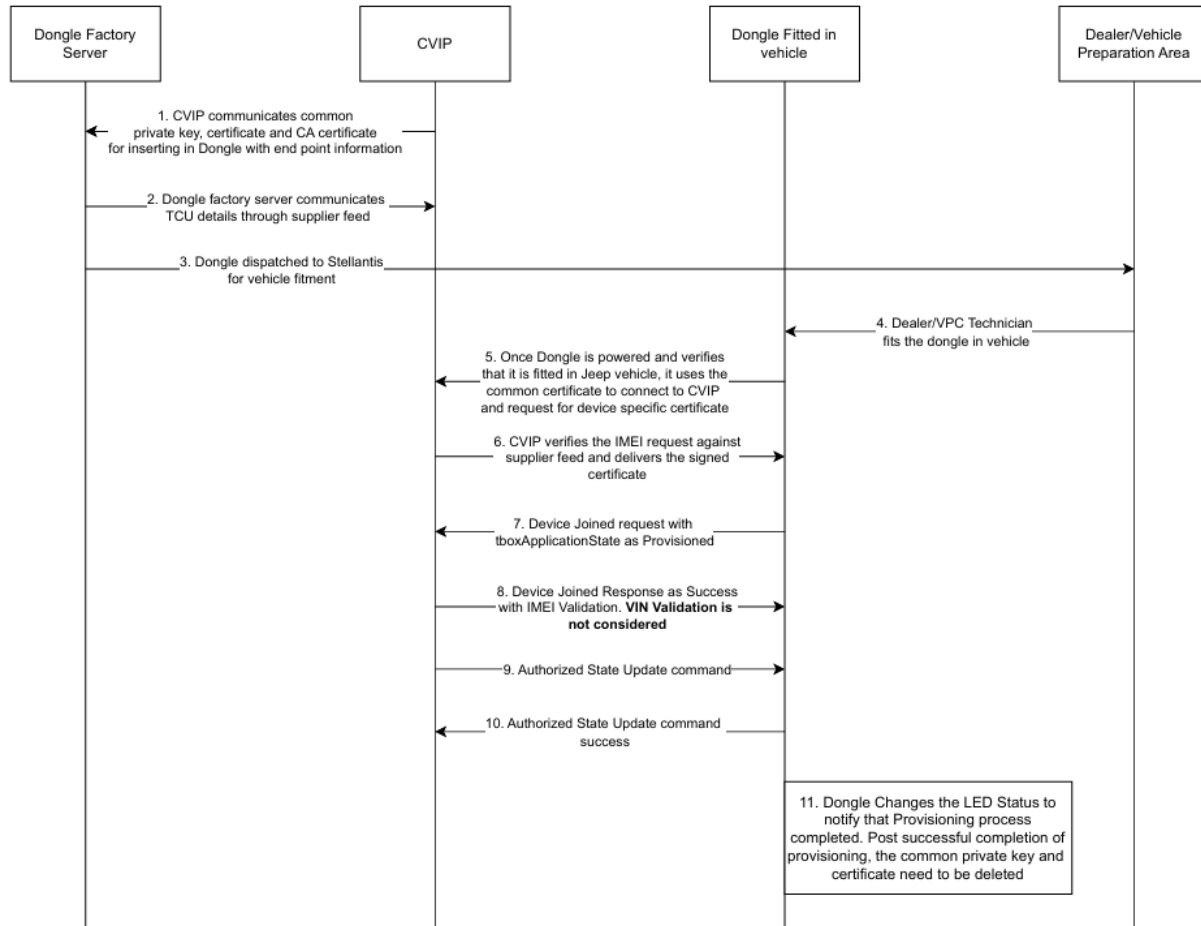


Figure 4: Vehicle Discovery Sequence

7.3.2 Vehicle Discovery Steps

- Dongle is powered up and TLS Session (TLS 1.2) is established.
- After that, DONGLE publishes message on the topic “/dongle/{VIN}/{protocol} {data format}/deviceJoined” to CVIP.
- On receiving the device Joined message, CVIP does IMEI validation and publishes message on the topic “/dongle/{VIN}/{protocol} {data format}/deviceJoinedRsp” to the DONGLE acknowledging that device discovery has successfully completed.
- CVIP shall send an Authorized state update command to the device to update the state of device.
- Device discovery shall have exponential retry logic implemented
- When device is disconnected then CVIP will be initiating CVIP_SMS_WAKE_UP API to MNO and MNO will send SMS to device
- Sms wakeup will include the following topic

<<TCUSerialNumber>>-JEEPCVIPSMSWakeupForRemoteOperations-
<<IMEI>>-<<Timestamp>>

- After receiving sms Dongle needs to verify sender_id and content of sms, if both are authenticated then Dongle needs to send deviceJoined message.
- SMS should be triggered only if the device is in the disconnected status on the CVIP and device has not sent a full sleep event.
- After receiving the SMS wakeup, Dongle need to communicate the SAME deviceJoined Payload in the topic mentioned below

/dongle/<vin>/MQTTPROTOBUF/remoteSMSWakeup

CVIP will provide the SAME deviceJoined response message in the below topic.

/dongle/<vin>/MQTTPROTOBUF/remoteSMSWakeupRsp

- SMS wakeup scenario will be applied only in Customer state.

7.3.2.1 Assumptions

Total SMS length should be 140 characters.

NOTE: The points in assumptions to be determined after discussion with Dongle Manufacturer and updated in the subsequent version of the document.

Exponential Retry Logic:

The exponential retry will take place in the below sequence,

<i>Steps</i>	<i>Retry Time</i>
<i>1</i>	<i>2s</i>
<i>2</i>	<i>4s</i>
<i>3</i>	<i>8s</i>
<i>4</i>	<i>16s</i>
<i>5</i>	<i>32s</i>
<i>6</i>	<i>64s</i>
<i>7</i>	<i>128s</i>
<i>8</i>	<i>SET DTC</i>

Step 1 to Step 8 will be iterated once in 10 minutes

This completes the vehicle discovery procedure.

Note : *The Dongle keeps retransmitting the deviceJoined message for exponential time until it receives a response from the CVIP cloud.*

7.3.3 MQTT Topics for Vehicle Discovery

Sno	Topic	Publisher	Subscriber	Description
1	/dongle/{VIN}/{protocol}{dataformat}/deviceJoined	Dongle	CVIP	Device will publish the device joined message
4	/dongle/{VIN}/{protocol}{dataformat}/deviceJoinedRsp	CVIP	Dongle	CVIP will publish deviceJoined Response

7.3.4 Message Formats

7.3.4.1 ICE

7.3.4.1.1 Message format for topic “/dongle/{VIN}/{protocol}{data format}/deviceJoined”

Parameter	Mandatory	Type	Description
vehicleId	M	String	Unique id of vehicle (VIN number)
TboxSerialNum	M	String	Dongle Serial number
imeiNo	M	String	IMEI Number
protocolVersion	M	String	Version of the protocol.
TboxOperatingState	M	String	Current operating state of Dongle.
TboxApplicationState	M	String	Current application state of Dongle. Refer to Section DONGLE states

TboxeSimState	M	String	Current eSIM state. Refer to Section DONGLE states
NAD_SW_Version	M	String	NAD firmware Version
MCU_SW_Version	M	String	MCU firmware Version
FOTAID	O	String	Message ID received for FOTA trigger (refer Section 10)
status	O	String	Status of FOTA installation (refer Section 10) DONGLEInstallationSuccess, DONGLEInstallationFailure]
releaseVersion	O	String	FOTA package release version (refer Section 10)

Sample topic: /dongle/WAUD2AFD7DN006931/MQTTPROTOBUF/deviceJoined

Payload Type: JSON

Sample Payload: {

```

    "vehicleId": "WAUD2AFD7DN006931",
    "TboxSerialNum": "SN29482029",
    "imeiNo": "490154203237518",
    "protocolVersion": "2.0.0",
    "TboxOperatingState": "NORMAL",
    "TboxApplicationState": "PROVISIONED",
    "TboxeSimState": "NORMAL_SIM",

    "NAD_SW_Version": "CD.02.03",
    "MCU_SW_Version": "VD0.02.03",
    "FOTAID": "dd52bab1-c64a-431b-a945-d88a80f74971",
    "status": "TBOXInstallationSuccess",
    "releaseVersion": "1.0"
  }

```

7.3.4.1.2 Message format for topic “/dongle/{VIN}/{protocol}}{data format}/deviceJoinedRsp”

Parameter	Mandatory	Type	Description
-----------	-----------	------	-------------

deviceJoinedRsp	M	String	Acknowledgement from from CVIP to Dongle
-----------------	---	--------	--

Sample topic: /dongle/WAUD2AFD7DN006931/MQTTPROTOBUF/deviceJoinedRsp

Payload Type: JSON

Sample Payload:

```
{
  "deviceJoinedAck": "Success"
}
```

7.3.4.1.3 Regex Parameter Validation for Device Joined Parameters

All Device joined payload needs to be UTF-8 validated to ensure response from CVIP.

Field name	Pattern/Allowed values
vehicleId	^[a-zA-Z0-9]{17,20}\$
TboxSerialNum	^[a-zA-Z0-9]{13}\$
protocolVersion	([0-9][.])([0-9][.])[0-9]
imeiNo	^[0-9]{15}\$
TboxOperatingState	["normal", "disconnected", "NORMAL", "DISCONNECTED"]
TboxApplicationState	factory, "FACTORY", "provisioned", "PROVISIONED", "authorized", "AUTHORIZED", "customer", "CUSTOMER"
TboxeSimState	"normal_sim", "NORMAL_SIM"
NADVersion	^[a-zA-Z0-9]{3}\.[a-zA-Z0-9]{2}\.[a-zA-Z0-9]{2}\$
MCUVersion	^[a-zA-Z0-9]{3}\.[a-zA-Z0-9]{2}\.[a-zA-Z0-9]{2}\$
FOTAID	^[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}\$
status	"TBOXInstallationSuccess", "TBOXInstallationFailure", "NADInstallationSuccess", "NADInstallationFailure"
releaseVersion	([0-9][.])[0-9]

7.3.5 FOTA Version Decoding:

SB API Interface ID : SB-DONGLE_ACTIVATION-008

Every software released will have a release version. The release version decoding logic is illustrated in the below diagram,

Dongle S/W Version Information

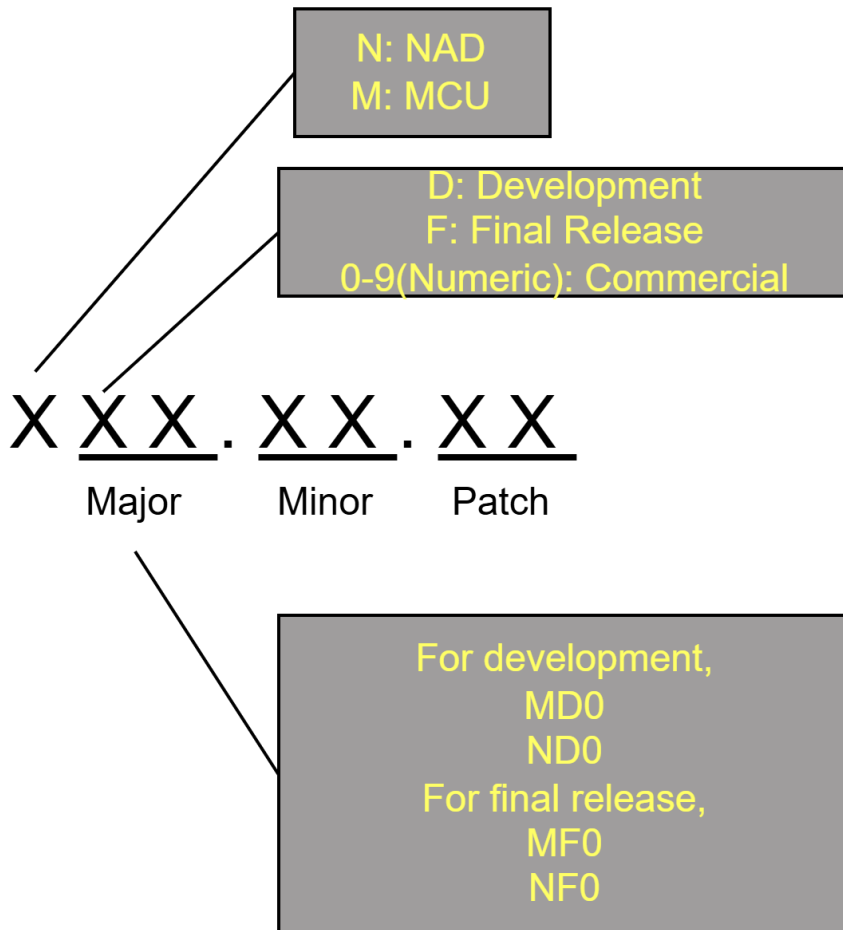


Figure 5: Fota Version Decoding

7.3.6 Device Joined – Corner Cases or Negative Scenarios Handling

- If internet is not connected during power on or MQTT session establishment, then Dongle will not queue up the Device joined message as per retry logic and communicate a burst of Device joined Message to CVIP server. Instead, Dongle shall wait till the Internet connectivity is established and only then initiate a device joined message.
- If Device joined failure DTC is set, then the Dongle shall not initiate a device joined retry.
- FOTA status parameter in device join message shall be applicable only in the following scenarios.
 - FOTA status parameters such as FOTAID, status, version would be added only in

- case FOTA update/install has been started.
- FOTA status parameters shall not be added in any device join message that happened before FOTA Update/Install starting.

8. Interface Definitions

Interface definitions are broadly categorized into two based on the mode of transmission

- Telemetry Streams – continuous stream of data
- Alerts & Events – based on Incident/occurrence of event

The following payload is common across telemetry streams, events and alerts

Message id	Unique identity for a message
TboxOperatingState	[normal, faulty, offline]
eTboxApplicationState	[factory, provisioned, customer]
eTboxeSimState	[bootstrap, normal_sim]
version	Protocol version for Jeep [value = 2.0.0]
time_stamp	Time at which the message is generated at the Dongle (epoch in milli sec)

The expected format of the timestamp is time since epoch in milli seconds.

The Dongle shall be synchronized with the Amazon Time Sync Service delivered over Network Time Protocol (NTP).

Dongle is expected to send data in the Big-Endian format.

8.1 Topic Nomenclature

SB API Interface ID : SB-INTERFACE_DEFINITION-001

Each message type will have respective topic endpoint as below,

Sr No	Message Type	Topic Endpoint Details	Vehicle Type
1	Vehicle	/dongle/{vinnum}/{protocol}{dataformat}/vehicleTelemetry	ICE

	Telemetry		
2	Location Telemetry	/dongle/{vinnum}/{protocol}{dataformat}/locationTelemetry	ICE
3	Vehicle Alerts	/dongle/{vinnum}/{protocol}{dataformat}/alerts	ICE
4	Vehicle Events	/dongle/{vinnum}/{protocol}{dataformat}/events	ICE
5	Emergency & Safety Alerts	/dongle/{vinnum}/{protocol}{dataformat}/emergencyAlerts	ICE
6	Diagnostic Alerts	/dongle/{vinnum}/{protocol}{dataformat}/diagnosticAlerts	ICE
7	Wakeup, Configuration command	/dongle/vinnumber/{protocol}{dataformat}/command	ICE
8	Wakeup, Configuration command response	/dongle/vinnumber/{protocol}{dataformat}/commandResponse	ICE
9	FOTA Command	/dongle/vinnumber/{protocol}{dataformat}/fotaCommand	ICE
10	FOTA Command Response	/dongle/vinnumber/{protocol}{dataformat}/fotaCommandResponse	ICE
11	TripStart	/dongle/{vinnum}/{protocol}{dataformat}/tripStart	ICE
12	TripCurrent	/dongle/{vinnum}/{protocol}{dataformat}/tripCurrent	ICE
13	TripEnd	/dongle/{vinnum}/{protocol}{dataformat}/tripEnd	ICE

8.2 Buffered data

SB API Interface ID : SB-INTERFACE_DEFINITION-002

Data accumulated upon connectivity failures are to be sent to the platform via separate topic upon network recovery.

Each message type will have respective topic endpoint for sending history data as below,

Sr No	Message Type	Topic Endpoint Details	Vehicle Type
1	Accumulated Vehicle Telemetry	/dongle/{vinnum}/{protocol}/{}/vehicleTelemetry/history	ICE
2	Accumulated Location Telemetry	/dongle/{vinnum}/{protocol}{dataformat}/locationTelemetry/history	ICE
3	Accumulated Vehicle Alerts	/dongle/{vinnum}/{protocol}{dataformat}/alerts/history	ICE
4	Accumulated Vehicle Events	/dongle/{vinnum}/{protocol}{dataformat}/events/history	ICE
5	Accumulated Emergency & Safety Alerts	/dongle/{vinnum}/{protocol}{dataformat}/emergencyAlerts/history	ICE
6	Accumulated Diagnostic Alerts	/dongle/{vinnum}/{protocol}{dataformat}/diagnosticAlerts/history	ICE
7	Accumulated Trip Start	/dongle/{vinnum}/{protocol}{dataformat}/tripStart/history	ICE
8	Accumulated Trip Progress	/dongle/{vinnum}/{protocol}{dataformat}/tripCurrent/history	ICE
9	Accumulated Trip End	/dongle/{vinnum}/{protocol}{dataformat}/tripEnd/history	ICE

9. DONGLE Commands

9.1 MQTT topics for wakeup and configuration command:

SB API Interface ID : SB-DONGLE_COMMANDS-001

S. no	Topic	Publisher	Subscriber	Description
1	/dongle/vinnumber/{protocol}{dataformat}/command	CVIP	Dongle	CVIP will send commands to Dongle

2	/dongle/vinnumber/{protocol} {dataformat}/commandResponse	Dongle	CVIP	Dongle will respond to CVIP with a Response
---	---	--------	------	---

9.2 SMS Wake up request:

SB API Interface ID : SB-DONGLE_COMMANDS-002

NOTE_2: For data required during SMS wakeup command please refer section 16.1

When the ignition is on, the data gets reflected automatically on the application. However, when the ignition is turned off

- SMS wakeup is triggered to dongle.
- Dongle wakes up and establishes MQTT session and publishes the Remote sms wake up MQTT payload
- CVIP responds to the remote sms wake up payload
- CVIP sends the command which is queued.
- DONGLE processes the command and goes back to sleep

9.3 Fetch Device Logs:

SB API Interface ID : SB-DONGLE_COMMANDS-003

In order to fetch the device logs associated with Dongle, CVIP triggers a command to Dongle. After the command is triggered from CVIP to Dongle, Dongle responds back with an acknowledgement and uploads the device logs using the URL provided in the FetchLogCommand (for example https://Dongle_debugger/deviceLogs/).

9.4 File Upload Failure error codes :

Sno	Error Codes	Description
1	invalidErrorCode(00)	File upload using secure URL à success
2	fileUploadFailure_accessTokenExpired(01)	File upload failure as Access token expired
3	fileUploadFailure_networkFailure(02)	File upload failure due to unavailable network
4	fileUploadFailure_urlError(03)	File upload failure due to url error

Sample Payload:

message_id: "dd52bab1-c64a-431b-a945-d88a80f74971"

```
Tbox_operating_state: normal
e_Tbox_application_state: provisioned
Tbox_esim_state: normal_sim
version: "2.0.0"
time_stamp {
  seconds: 1626101887
  nanos: 947815000
}
fetchDeviceLogs {
  fetchDeviceLogsPayload {
    url: "https://Tbox_debugger/deviceLogs/"
    accessToken: "4e244668-f7ae-38d2-b8a7-d24ba3991144"
  }
}
```

Note: Device Fetch Log response to be sent after Log file uploaded using the secure URL provided.

Sample Response:

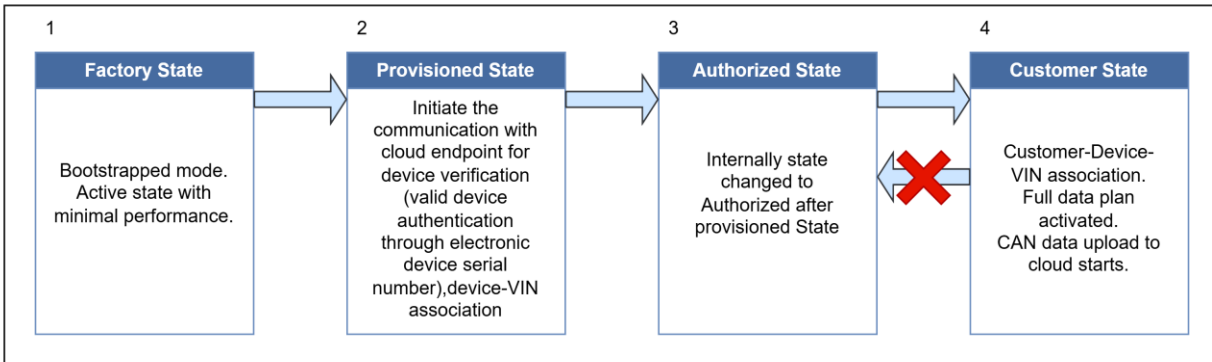
```
message_id: "dd52bab1-c64a-431b-a945-4910ab6ca7b3"
correlation_id: " dd52bab1-c64a-431b-a945-d88a80f74971"
e_Tbox_application_state: provisioned
Tbox_operating_state: normal
Tbox_esim_state: normal_sim
version: "2.0.0"
time_stamp {
  seconds: 1626101887
  nanos: 953214000
}
return_code: succeeded
deviceFetchLogsResponsePayload {
  deviceFetchLogsCommandResponse {
    fileUploadStatus: successful/failure
    errorCode: invalidErrorCode(00)/ fileUploadFailure_accessTokenExpired(01)/
fileUploadFailure_networkFailure(02)/ fileUploadFailure_urlError(03)
  }
}
```

9.5 Dongle State Update Command:

SB API Interface ID : SB-DONGLE_COMMANDS-004

Dongle can be moved to Authorized, Customer state using the state update command.

Below are the illustrations for Dongle state updates,



Sample Command Payload:

```

message_id: "45ee3134-8c28-4735-b208-c1g8bae52534"
subtype: TBOXStateUpdateCommand
version: "2.0.0"
time_stamp {
seconds: 1634302047
}
command_payload {
TboxStateUpdateCommandPayload {
TboxState: "CUSTOMER"
}
}
  
```

Sample Response Payload:

```

message_id: "dd52bab1-c64a-431b-a945-1602f26e9f4e"
correlation_id: "45ee3134-8c28-4735-b208-c1g8bae52534"
subtype: TBOXStateUpdateCommandResponse
Tbox_operating_state: faulty
e_Tbox_application_state: provisioned
Tbox_esim_state: normal_sim
  
```

```
version: "2.0.0"  
time_stamp {  
seconds: 1634136148  
nanos: 594897000  
}  
commandResponsePayload {  
TboxStateUpdateCommandResponsePayload {  
commandStatus: Success  
currentTboxState: CUSTOMER  
}  
}
```

9.6 User Defined Speed Alert Threshold Command:

SB API Interface ID : SB-DONGLE_COMMANDS-005

If the user intends to change the speed alert threshold in the mobile app, the user will change the overspeed threshold and communicate as a command to Dongle. The dongle need to consider the speed threshold with the value communicated in the command.

After setting the speed threshold, Dongle needs to send the command response payload as success and the set threshold speed.

Sample Command Payload:

```
message_id: "45ee3134-8c28-4735-b208-c1g8bae52534"  
subtype: UserDefinedSpeedCommand  
version: "2.0.0"  
time_stamp {  
seconds: 1634302047  
}  
command_payload {  
UserDefinedSpeedCommandPayload {  
speed: "95"  
}  
}
```

Sample Command Response payload:

```
message_id: "dd52bab1-c64a-431b-a945-4910ab6ca7b3"
correlation_id: "45ee3134-8c28-4735-b208-c1g8bae52534"
subtype: UserDefinedSpeedCommandResponse
e_Tbox_application_state: customer
Tbox_operating_state: normal
Tbox_esim_state: normal_sim
version: "2.0.0"
time_stamp {
  seconds: 1626101887
  nanos: 953214000
}
return_code: succeeded
commandResponsePayload {
  UserDefinedSpeedCommandResponse {
    commandStatus: success
    definedSpeed: "95"
  }
}
```

9.7 User Defined Minimum Trip Distance Command:

SB API Interface ID : SB-DONGLE_COMMANDS-006

If the user intends to change the minimum trip distance threshold in the mobile app, the user will change the minimum trip distance threshold and communicate as a command to Dongle. Dongle need to consider the minimum trip distance threshold with the value communicated in the command.

After setting the minimum trip distance threshold, Dongle need to send the command response payload as success and the set threshold speed.

Sample Command Payload:

```
message_id: "45ee3134-8c28-4735-b208-c1g8bae52534"
subtype: UserDefinedMinimumTripDistanceCommand
version: "2.0.0"
time_stamp {
  seconds: 1634302047
}
```

```
command_payload {  
  UserDefinedMinimumTripDistanceCommandPayload {  
    minimumTripDistance: "5",  
    engineOffTime:"15"  
  }  
}
```

Sample Command Response payload:

```
message_id: "dd52bab1-c64a-431b-a945-4910ab6ca7b3"  
correlation_id: "45ee3134-8c28-4735-b208-c1g8bae52534"  
subtype: UserDefinedMinimumTripDistanceCommandResponse  
e_Tbox_application_state: customer  
Tbox_operating_state: normal  
Tbox_esim_state: normal_sim  
version: "2.0.0"  
time_stamp {  
  seconds: 1626101887  
  nanos: 953214000  
}  
return_code: succeeded  
commandResponsePayload {  
  UserDefinedMinimumTripDistanceCommandResponse {  
    commandStatus: success  
    definedMinimumTripDistance: "5",  
    DefinedEngineOffTime : "15"  
  }  
}
```

9.8 Key Design Considerations for Remote Operations

If the command is received by Dongle after 60 seconds of command-initiated timestamp, such commands need to be ignored by Dongle and need not to be processed.

60s timeout duration need to be further fine-tuned.

When ever Dongle receives remote command from CVIP, Dongle needs to send command response as well as event payload.

9.9 Remote Blinker On Command:

SB API Interface ID : SB-REMOTE_COMMANDS-001

If the user intends to get Remote Blinker On command in the mobile app, the user will change the blinker command and communicate as a command to Dongle. Dongle need to consider blinker command with the value communicated in the command.

After setting the blinker command, Dongle need to send the command response payload as success ,if Remote blinker activation has passed.

Dongle need to send the command response payload as fail, if Remote blinker activation has failed.

Sample Command Payload:

message_id: "dd52bab1-c64a-431b-a945-e57ea908f6b7"

subtype: RemoteBlinkerOnCommand

version: "2.0.0"

time_stamp {

seconds: 1682671998

nanos: 391110000

}

command_payload {

remoteBlinkerOnCommandPayload {

BlinkerOnEventStatus: On

remoteBlinkerOnTimestamp {

seconds: 1682671998

nanos: 391216000

}

}

}

data published

Sample Command Response payload:

subtype: RemoteBlinkerOnCommandResponse

Tbox_operating_state: disconnected

e_Tbox_application_state: provisioned

Tbox_esim_state: normal_sim

version: "2.0.0"

time_stamp {

seconds: 1679891262

```

    nanos: 380019000
  }
  commandResponsePayload {
    remoteBlinkerOnCommandResponsePayload {
      commandStatus: failure
    }
  }
}

```

9.10 Remote Blinker Off Command:

SB API Interface ID : SB-REMOTE_COMMANDS-002

If the user intends to get Remote Blinker Off command in the mobile app, the user will change the blinker command and communicate as a command to Dongle. Dongle need to consider blinker command with the value communicated in the command.

After setting the blinkers command, Dongle need to send the command response payload as success ,if Remote blinker deactivation has passed.

Dongle need to send the command response payload as fail, if Remote blinker deactivation has failed.

Sample Command Payload:

```

message_id: "dd52bab1-c64a-431b-a945-e57ea908f6b7"
subtype: RemoteBlinkerOffCommand
version: "2.0.0"
time_stamp {
  seconds: 1682671998
  nanos: 391110000
}
command_payload {
  remoteBlinkerOffCommandPayload {
    BlinkerOffEventStatus: Off
    remoteBlinkerOffTimestamp {
      seconds: 1682671998
      nanos: 391216000
    }
  }
}
}

```

data published

Sample Command Response payload:

```
subtype: RemoteBlinkerOffCommandResponse
Tbox_operating_state: disconnected
e_Tbox_application_state: provisioned
Tbox_esim_state: normal_sim
version: "2.0.0"
time_stamp {
  seconds: 1679891262
  nanos: 380019000
}
commandResponsePayload {
  remoteBlinkerOffCommandResponsePayload {
    commandStatus: failure
  }
}
```

9.11 Remote Door Lock Command

SB API Interface ID : SB-REMOTE_COMMANDS-003

If the user intends to get Remote Door Lock command in the mobile app, the user will change the remote door lock command and communicate as a command to Dongle. Dongle need to consider remote door lock command with the value communicated in the command.

After setting the remote door lock command, Dongle need to send the command response payload as success.

Dongle need to send the command response payload as fail, if Remote lock activation has failed.

Sample Command payload:

```
message_id: "dd52bab1-c64a-431b-a945-4846fcd10343"
subtype: RemoteDoorLockCommand
version: "2.0.0"
time_stamp {
  seconds: 1679891142
  nanos: 74428000
}
```

```

}
command_payload {
  remoteDoorLockCommandPayload {
doorLockTimeStamp{
  seconds: 1682671998
  nanos: 391216000
}

}
}
}
Sample Command Response payload:
message_id: "dd52bab1-c64a-431b-a945-e253dd94094d"
subtype: RemoteDoorLockCommandResponse
Tbox_operating_state: disconnected
e_Tbox_application_state: provisioned
Tbox_esim_state: normal_sim
version: "2.0.0"
time_stamp {
  seconds: 1679891422
  nanos: 832406000
}
commandResponsePayload {
  remoteDoorLockCommandResponsePayload {
    commandStatus: success
  }
}

```

9.12 Remote Door UnLock Command

SB API Interface ID : SB-REMOTE_COMMANDS-004

If the user intends to get Remote Door UnLock command in the mobile app, the user will change the remote door unlock command and communicate as a command to Dongle. Dongle need to consider remote door unlock command with the value communicated in the command.

After setting the remote door unlock command, Dongle need to send the command response payload as success

Dongle need to send the command response payload as fail, if Remote unlock activation has failed.

Sample Command payload

```
message_id: "dd52bab1-c64a-431b-a945-1ba1587ac32c"
subtype: RemoteDoorUnLockCommand
version: "2.0.0"
time_stamp {
  seconds: 1679891563
  nanos: 527834000
}
command_payload {
  remoteDoorLockCommandPayload {
    commandStatus: success
  }
  doorUnLockTimeStamp{
    seconds: 1682671998
    nanos: 391216000
  }
}
}
```

Sample Command Response payload:

```
message_id: "dd52bab1-c64a-431b-a945-526c46668bb0"
subtype: RemoteDoorUnLockCommandResponse
Tbox_operating_state: disconnected
e_Tbox_application_state: provisioned
Tbox_esim_state: normal_sim
version: "2.0.0"
time_stamp {
  seconds: 1679891983
  nanos: 254412000
}
commandResponsePayload {
  remoteDoorUnLockCommandResponsePayload {
    commandStatus: success
  }
}
```

9.13 Remote Honk Command

SB API Interface ID : SB-REMOTE_COMMANDS-005

If the user intends to get Remote Honk command in the mobile app, the user will change the remote Honk command and communicate as a command to Dongle. Dongle need to consider remote honk command with the value communicated in the command.

After setting the remote honk command, Dongle need to send the command response payload as success

Dongle need to send the command response payload as fail, if Remote honk activation has failed.

Sample Command payload

message_id: "dd52bab1-c64a-431b-a945-1ba1587ac32c"

subtype: RemoteHonkCommand

version: "2.0.0"

time_stamp {

seconds: 1679891563

nanos: 527834000

}

command_payload {

remoteHonkCommandPayload {

TimeStamp{

seconds: 1682671998

nanos: 391216000

}

}

}

Sample Command Response payload:

message_id: "dd52bab1-c64a-431b-a945-526c46668bb0"

subtype: RemoteHonkCommandResponse

Tbox_operating_state: disconnected

e_Tbox_application_state: provisioned

Tbox_esim_state: normal_sim

version: "2.0.0"

time_stamp {

seconds: 1679891983

nanos: 254412000

}

commandResponsePayload {

remoteHonkCommandResponsePayload {

remoteHonkFailureCode: requestFailed

```
}  
}
```

9.14 Remote Commands Failure Codes

SB API Interface ID : SB-REMOTE_COMMANDS-006

NOTE: The Remote command Failure Codes will be determined after discussion with DONGLE MANUFACTURER and will be updated/modified in next version of the SB interface document.

Following remote command failure codes shall be communicated by Dongle to CVIP with Proto enumeration.

- KeyIn_ACC = 1;
- KeyIn_IGN = 2;
- Engine_Running_or_Ready_ToDrive = 3;
- LVBAT_LowVoltage_or_High_Voltage = 4;
- Limp_Home_Mode_or_Reserved = 5;
- Crashed_Detected_ = 6;
- RequestRefused_DoorOpen = 7;
- Bounce=8;
- FotalInProgress=9;
- Cranking =10;
- CrankingFailedDueTo_DoorOpen_or_Unlock=11;
- CrankingFailedDueTo_Not_In_ParkPosition=12;
- CrankingFailedDueTo_PWT_or_VehicleFailure=13;
- ACSwitchNotON=14;
- UnUsedPreConditioning=15;
- authentication_failed=16;
- authentication_timeout=17;
- request_failed=18;
- response_timeOut=19;
- rCDOOnFailure=20;
- rCDOOnBCCMNotSending432=21;
- commandFailedDTCSet=22;
- DoorOpenLock=23;
- DoorsAutoLockedAfterTimeout=24;
- RequestRefused=25;
- RequestFailed=26;
- CrankingFailedDueToLowFuel=27;
- CompressorFailure=28;

9.15 Remote SMS Wakeup

SB API Interface ID : SB-REMOTE_COMMANDS-007

9.15.1 KeyConsiderations

- When device is in DISCONNECTED state , SMS wakeup is triggered from CVIP
- Upon receiving the SMS, DONGLE wakes up and sends message in remoteSMSWakeup topic.
- CVIP responds back with remoteSMSWakeupRsp ,post which CVIP will trigger the pending command.
- DONGLE will take need action for the command received as per usual flow.

9.15.2 Regex Parameter Validation for Remote SMS Wakeup Parameters

All remoteSMSWakeup payload needs to be UTF-8 validated to ensure response from CVIP.

Field name	Pattern/Allowed values
vehicleId	^[a-zA-Z0-9]{17,20}\$
TboxSerialNum	^[a-zA-Z0-9]{13}\$
protocolVersion	([0-9][.])([0-9][.])[0-9]
imeiNo	^[0-9]{15}\$
TboxOperatingState	["normal", "disconnected", "NORMAL", "DISCONNECTED"]
TboxApplicationState	factory, "FACTORY", "provisioned", "PROVISIONED", "customer", "CUSTOMER"
TboxeSimState	"normal_sim", "NORMAL_SIM"
NADVersion	^[a-zA-Z0-9]{3}\.[a-zA-Z0-9]{2}\.[a-zA-Z0-9]{2}\$
MCUVersion	^[a-zA-Z0-9]{3}\.[a-zA-Z0-9]{2}\.[a-zA-Z0-9]{2}\$
FOTAID	^[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}\$
status	"TBOXInstallationSuccess", "TBOXInstallationFailure", "NADInstallationSuccess", "NADInstallationFailure"
releaseVersion	([0-9][.])[0-9]

9.15.3 Message format for topic “/dongle/{VIN}/{protocol}{data format}/remoteSMSWakeup”

Parameter	Mandatory	Type	Description
vehicleId	M	String	Unique id of vehicle (VIN number)
TboxSerialNum	M	String	Dongle Serial number
imeiNo	M	String	IMEI Number

protocolVersion	M	String	Version of the protocol.
TboxOperatingState	M	String	Current operating state of Dongle.
TboxApplicationState	M	String	Current application state of Dongle. Refer to Section DONGLE states
TboxeSimState	M	String	Current eSIM state. Refer to Section DONGLE states
NADVersion	M	String	NAD firmware Version
MCUVersion	M	String	MCU firmware Version

Sample topic: /dongle/WAUD2AFD7DN006931/MQTTPROTOBUF/
remoteSMSWakeup

Payload Type: JSON

Sample Payload: {
 "vehicleId": "WAUD2AFD7DN006931",
 "TboxSerialNum": "SN29482029",
 "imeiNo": "490154203237518",
 "protocolVersion": "2.0.0",
 "TboxOperatingState": "NORMAL",
 "TboxApplicationState": "CUSTOMER", "TboxeSimState": "NORMAL_SIM",
 "NADVersion": "CD.02.03",
 "MCUVersion": "VD0.02.03"
}

9.15.4 Message format for topic “/dongle/{VIN}/{protocol}}{data format}/remoteSMSWakeupRsp”

Parameter	Mandatory	Type	Description
deviceJoinedAck	M	String	Acknowledgement from from CVIP to Dongle

Sample topic: /dongle/WAUD2AFD7DN006931/MQTTPROTOBUF/
remoteSMSWakeupRsp

Payload Type: JSON

Sample Payload:

```
{  
  "deviceJoinedAck": "Success"  
}
```

10. FOTA – Firmware over the air

10.1 MQTT topics for FOTA command and command response:

SB API Interface ID : SB-FOTA_COMMANDS-001

S. no	Topic	Publisher	Subscriber	Description
1	/dongle/vinnumber/{protocol}{dataformat}/fotaCommand	CVIP	Dongle	CVIP will send commands to Dongle
2	/dongle/vinnumber/{protocol}{dataformat}/fotaCommandResponse	Dongle	CVIP	Dongle will respond to CVIP with a Response

10.2 Remote Firmware Check Command

SB API Interface ID : SB-FOTA_COMMANDS-002

This Command is used to check the current firmware version of the Dongle at any given point of time and doesn't have any linkage with FOTA workflow.

Sample Command Payload:

message_id: "dd52bab1-c64a-431b-a945-d88a80f74971"

subtype: RemoteFirmwareCheckCommand

version: "2.0.0"

time_stamp {

seconds: 1626780137

nanos: 510125000

}

fota_command_payload {

remoteFirmwareCheckCommand {

ecuCode: TBOX

}

}

Sample Response Payload :

message_id: "dd52bab1-c64a-431b-a945-1ce46ccc9a70"

correlation_id: " dd52bab1-c64a-431b-a945-d88a80f74971"

subtype: RemoteFirmwareCheckCommandResponse

```
Tbox_operating_state: normal
e_Tbox_application_state: provisioned
Tbox_esim_state: normal_sim
version: "2.0.0"
time_stamp {
  seconds: 1626780192
  nanos: 504139000
}
return_code: fota_succeeded
fotacommandResponsePayload {
  remoteFirmwareCheckCommandResponse {
    ecuCode: TBOX
    NADVersion: "Version 2.0"
    MCUVersion: "Version 2.0"
  }
}
```

10.3 Firmware Upload API

SB API Interface ID : SB-FOTA_COMMANDS-003

Firmware package to be uploaded by OEM to the CVIP platform through the run management portal. The image is expected to be tested thoroughly before uploading to the CVIP production environment.

10.3.1 Design Consideration

- The image shall be uploaded only by an authorized stakeholder from OEM.
- The image may be a MCU image file or a NAD image file.
- The image shall always include a metadata.json file which shall mandatorily include the following parameter
 - o Release Version
 - o MCU Version
 - o NAD Version
 - o Compatible Release Version
 - o Compatible NAD Version
 - o Compatible MCU Version

- o Whether the image is golden or not

- o Image Size

- o SHA-256 Checksum

- The platform shall also verify the actual image size with the size mentioned in metadata.
- The SHA-256 checksum needs to be validated in the platform and the upload should be succeeded only upon the checksum matching
- The platform shall not accept the image upload if the image size verification or SHA-256 checksum validation fails
- The platform shall allow the administrator to update or delete a specific image. However, the image which is getting updated or deleted shall not be associated with any campaign.
- All the release version shall be unique and strictly incremental. The platform shall not allow the image upload which is less than the latest image which is available in platform.
- The release version and the MCU/NAD version shall be unique. The platform shall not allow duplicate NAD/MCU version
- The file format shall be mutually agreed between OEM and Platform team. The platform shall allow only the file format which is agreed upon.
- Post upload, the platform shall have the provision to display the entire list of firmware along with the following attributes.

- o Release Version

- o NAD Version

- o MCU Version

- o Golden Image applicability

- o Image Size

- o Date & time of upload

- o Association with any Campaign

- o Status

- The platform shall have the feasibility to search firmware based on the name or release version or date of upload or period of upload

Sample Payload:

```
[{
  "type": "mcu",
  "firmwareInstallType": "FULL",
  "updateVersion": "2.0",
  "file": "TBOX-MCU.XXX",
  "softwareSize": "102920",
  "downloadHash": "ba7816bf8f01cfea414140de5dae2223b00361a396177a9cb410ff61f20016ab",
  "encryptionHash": "ba7816bf8f01cfea414140de5dae2223b00361a396177a9cb410ff61f20016ab"
},{
  "type": "nad",
  "firmwareInstallType": "FULL",
  "updateVersion": "2.0",
  "file": "TBOX-NAD.XXX",
  "softwareSize": "102920",
  "downloadHash": "ba7816bf8f01cfea414140de5dae2223b00361a396177a9cb410ff61f20016ab",
  "encryptionHash": "ba7816bf8f01cfea414140de5dae2223b00361a396177a9cb410ff61f20016ab"
}]
```

10.4 Firmware Download Command:

SB API Interface ID : SB-FOTA_COMMANDS-004

NOTE: The firmware types will be determined after discussion with DONGLE MANUFACTURER and will be added in next version of the SB interface document.

After firmware download command is triggered from CVIP, Dongle acknowledges the FOTA command with a response. In case the FOTA command was received in IGN ON, DONGLE will

save it and start update/install in IGN OFF. In IGN OFF, DONGLE start update/install procedure after trip end(2h).

If Dongle is in Authorized state, the following requirements shall be considered.

- If IGN OFF, DONGLE cannot receive the FOTA command because DONGLE is in deep sleep mode.
- If IGN ON, DONGLE can receive the FOTA command.
- If IGN changed to ON from OFF while FOTA in progress, DONGLE keeps the FOTA progress without stop.

A firmware download command resembles like this

```
message_id: "ecf43b70-11c2-11ed-a31d-47ed1f41b941"
subtype: FirmwareDownloadCommand
version: "2.0.0"
time_stamp {
  seconds: 1659376446
}
fota_command_payload {
  firmwareDownloadCommand {
    firmwareDownloadPayload {
      firmwareType: mcu
      url:
"https://65.0.235.157:50545/files/device/downloadfirmware?filename=VT0.09.00.enc&releaseve
rsion=5.0&filetype=mcu&category=BATCH"
      accessToken: "b34495b1-ba6d-36d8-a2cd-43b8db03aee6"
      checkSum: "3057321fc9ec80fbb669d028e715c96793a2672c17824064cfb882497b8a2911"
      rawCheckSum:
"3057321fc9ec80fbb669d028e715c96793a2672c17824064cfb882497b8a2911"
      softwareSize: "1036672"
      installType: Full
      releaseVersion: "5.0"
      rawSize: "1036672"
      signature:
"laFPPdsrVIESgYYVvY9wxji9MaMKw2IYB3+bYHeGQ6CmVunkymVd5IMTzBI/cjIBPDvrJ
0kdq2rgb9ah/CeQnzUXfdAOTnOrXsYVhXlQ0VnQ/0sVkp4K8y8A8XhW4oJX196tjQooP/C/ez
5SOAq2TI1kBqEkiaMRQ116G4xjhIWAM2PaJp/qpc7o0uE76RTljBXTJTHG9SyjB4x0TqVPT
DSpuwbrxKK+haVy/sqigedNZTDzt2R7VRwC6LfWL52aBRcd35mV1jVKymK1PEriSWuytzZ
G9FIEgfQoZkguJbECnPhU1tZTio8bsan/AZFLTGct6uVT0+OICaL4+g0p2A=="
    }
    firmwareDownloadPayload {
```

```

    firmwareType: nad
    url:
"https://65.0.235.157:50545/files/device/downloadfirmware?filename=CT0.09.00.enc&releaseve
rsion=5.0&filetype=nad&category=BATCH"
    accessToken: "b34495b1-ba6d-36d8-a2cd-43b8db03aee6"
    checksum: "27639cde9e90eaa5304278e2d3c9ebb1c8e9a2878474ac4cb5f9922c04ba7e01"
    rawChecksum:
"27639cde9e90eaa5304278e2d3c9ebb1c8e9a2878474ac4cb5f9922c04ba7e01"
    softwareSize: "77306448"
    installType: Full
    releaseVersion: "5.0"
    rawSize: "77306448"
    signature:
"NGIj7h65hQvnjySWGourtgs8tP2Rhjhj6cXzcn9xQ8FIOsQDqLCKPvdploEHZBcO4Itta2JIoaY
KCd0RxYx6hBE1F9arHehw2eBxI51Doy2bL2NrZrF6avzLEt1q6DjhL8m4KNzubs0XEt6FPzbv
IIXEj5uxMSKQ6t6q7WP1oLodiDWOmNC+WQLLa2g/GAtkUX9fGcTGMOKM8rnrLN5uTX1
GYLqjx1Du3GtYbR9gD2T5/y17OqBX2zyveD8FEZpEjHG4DbLOQMTOT21CBWYZKsxF/4Q
H5bPENPDcb7mPqrl8+aBdJ3NYmdi/c882Vf/SAob6Wff3Mganh+siyqDxcA=="
    }
  }
}

```

The command includes the following parameter which is explained in detail.

- Command ID or message_id
This UUID is supposed to be passed as an input parameter during the download request while hitting the URL. Also, this parameter is supposed to be communicated as FOTAID as part of device joined post device reboot after FOTA installation.
- firmwareType
This parameter specifies if the package is applicable for MCU or NAD
- URL
The download URL which needs to be accessed by the Dongle to initiate the download.
- accessToken
Authentication token which needs to be passed as authorization header during download initiation. This token is valid for a maximum duration of 7 days.
- Checksum
SHA-256 checksum of the FOTA package to validate the hash values of encrypted FOTA package
- rawChecksum
SHA-256 checksum of the FOTA package to validate the hash values of un-encrypted FOTA package
- SoftwareSize
Size of the encrypted FOTA package in bytes.

- **rawSize**
Size of the un-encrypted FOTA package in bytes.
- **Signature**
Signature value of the encrypted FOTA package.

Upon receiving the FOTA command, the Dongle shall execute the following steps.

- After downloading the package, the Dongle shall verify the signature of the FOTA package using the signing certificate
 - If the signature verification is successful, then Dongle shall initiate the FOTA package decryption
 - If the signature version fails, then the Dongle shall communicate the error `DecodeMCUSignatureAuthenticationFailure` or `DecodeNADSignatureAuthenticationFailure`
- After signature verification, the Dongle shall verify the file size mentioned in the FOTA package matches with the actual encrypted FOTA package.
 - If the file size matches, then Dongle can proceed to next step
 - If the file size mismatches, then the Dongle shall send the error code `DownloadInProgressNADFilesizeNomatched` or `DownloadInProgressVMPUFilesizeNomatched`
- If the file size matches, then Dongle shall use the encryption key to decrypt the FOTA package.
 - If the decryption is successful, then Dongle can proceed to next step.
 - If the decryption fails, then Dongle shall communicate `DecodeMCUWrongEncryptionFile` or `DecodeNADWrongEncryptionFile`
- If the decryption is successful, then Dongle shall verify the decrypted file size need to be verified with the `rawSize` parameter in FOTA download command.
- Dongle can initiate the FOTA installation post to these checks.

Dongle continuously updates the CVIP about its current state with the below error-codes.

10.4.1 FOTA

return

codes:

SB API Interface ID : SB-FOTA_COMMANDS-005

S. no	FOTA return codes	Description
1	Unknown Return code(0)	Unknown return code
2	FOTA succeeded (1)	FOTA update Successful
3	FOTA failed (2)	FOTA update failure

4	FOTA update (3)	FOTA update in progress
5	FOTA waiting (4)	FOTA waiting

10.4.2 Firmware download and installation response codes:

SB API Interface ID : SB-FOTA_COMMANDS-006

NOTE: The response codes will be determined after discussion with DONGLE MANUFACTURER and will be added in next version of the SB interface document.

After NAD install device is expected to reboot and communicate the status of FOTA in deviceJoined payload (Refer section 7.3 – Vehicle discovery)

S. no	Download and Install state	Description
1	Default (0)	Default state
2	MCU Download in Progress (1)	MCU Firmware download in progress
3	NAD Download in Progress (2)	NAD Firmware download in progress
4	MCU Download Successful(3)	MCU Firmware download successful
5	NAD Download Successful(4)	NAD Firmware download successful
6	MCU Ready to Install (5)	MCU Firmware download completed and ready to install
7	NAD Ready to Install (6)	NAD Firmware download completed and ready to install
8	MCU Installation in Progress (7)	MCU firmware installation in progress
9	Installation successful MCU(10)	MCU firmware installation success

10	CheckTboxCondition(11)	Dongle Condition Check (FOTA Package Version)
11	FOTAWaiting(12)	Waiting available time for FOTA Update and installation
12	CheckTboxCondition2(13)	Dongle Condition Check (Vehicle voltage and status)

10.4.3 FOTA Error codes:

SB API Interface ID : SB-FOTA_COMMANDS-007

NOTE: *The FOTA error codes will be determined after discussion with DONGLE MANUFACTURER and will be added in next version of the SB interface document.*

S. no	Error Codes	Description
1	InvalidFailureCode (0)	Invalid Failure Code
2	DownloadInProgressMCUDownloadRetryFailure(100)	MCU download failure
3	DownloadInProgressNADDownloadRetryFailure(101)	NAD download failure
4	DownloadInProgressMCUDownloadTimeout(102)	MCU download timeout
5	DownloadInProgressNADDownloadTimeout(103)	NAD download timeout
6	DownloadInProgressMCUFileValidationFailure(104)	MCU file validation failure (Checksum failure)
7	DownloadInProgressNADFileValidationFailure(105)	NAD file validation failure (Checksum failure)
8	DownloadInProgressMCUFilesizeNomatched(106)	MCU Software file size is not matching
9	DownloadInProgressNADFilesizeNomatched(107)	NAD Software file size is not matching
10	DownloadInProgressInvalidWrongPayload(108)	Wrong payload

11	DownloadMCUHTTPCommonError(111)	MCU invalid access code
12	DownloadNADHTTPCommonError(112)	NAD invalid access code
13	CheckTboxConditionLowVoltage(113)	Low voltage Condition
14	CheckTboxConditionInvalidGoldenImage(114)	Internal Golden Image Error
15	CheckTboxConditionInvalidMCUVersion(115)	Invalid MCU Version
16	CheckTboxConditionInvalidNADVersion(116)	Invalid NAD Version
17	CheckTboxConditionCommonError(117)	Other Error
18	DecodeMCUMismatchHash = 118	Decryption failed, mismatched hash
19	DecodeNADMismatchHash = 119	Decryption failed, mismatched hash
20	DecodeMCUWrongEncryptionFile = 120	Encryption file does not exist or cannot be opened
21	DecodeNADWrongEncryptionFile = 121	Encryption File does not exist or cannot be opened
22	DecodeMCUSignatureAuthenticationFailure = 122	Signature authentication failure
23	DecodeNADSignatureAuthenticationFailure = 123	Signature authentication failure

10.4.4 FOTA states and possible error codes:

SB API Interface ID : SB-FOTA_COMMANDS-008

S. no	Download and Install state	Possible Error codes
-------	----------------------------	----------------------

1	Default (0)	• InvalidFailureCode
2	MCU Download in Progress (1)	• MCUDownloadRetryFailure(100) • MCUDownloadTimeout(102) • MCUFileValidationFailure(104) • MCUFileSizeNomatched(106) • DownloadMCUHTTPCommonErr(111) • DecodeMCUMismatchHash(118) • DecodeMCUWrongEncryptionFile(120) • DecodeMCUSignatureAuthenticFailure(122)
3	NAD Download in Progress (2)	• NADDownloadRetryFailure(101) • NADDownloadTimeout(103) • NADFileValidationFailure(105) • NADFileSizeNomatched(107) • DownloadNADHTTPCommonErr(112) • DecodeNADMismatchHash(119) • DecodeNADWrongEncryptionFile(121) • DecodeNADSignatureAuthenticFailure(123)
4	MCU Download Successful(3)	• InvalidFailureCode
5	NAD Download Successful(4)	• InvalidFailureCode
6	MCU Ready to Install (5)	• InvalidFailureCode
7	NAD Ready to Install (6)	• InvalidFailureCode
8	MCU Installation in Progress (7)	• InvalidFailureCode
9	Installation successful MCU(10)	• InvalidFailureCode
10	Installation successful NAD(11)	• NAD firmware installation success
11	CheckTboxCondition(13)	• CheckTboxConditionLowVoltage (113) • CheckTboxConditionInvalidGoldenImage (114) • CheckTboxConditionInvalidMCUVersion (115)

		• CheckTboxConditionInvalidNADVersion (116)
--	--	--

11. Dongle Operating Modes

Data to be sent from Dongle to cloud can be categorized based on the operating mode of the vehicle

- Off
- Ignition On and Running

Note: Ignition states and signals transmitted are to be discussed with DONGLE MANUFACTURER and STELLANTIS ENGINEERING team

11.1 Off State

11.1.1.1 Full Sleep

In Full Sleep mode Dongle will not be able to communicate with CVIP.

Before moving to full sleep mode Dongle should communicate to CVIP about the state change.

Please refer section 13.2 vehicle events for the corresponding state change event.

11.2 Ignition on State

During the Ignition on state, the Dongle will be in normal mode and stays connected to the internet.

ICE Vehicles

In case of ICE, the Dongle sends the vehicle and location telemetry data based on the communication frequency. Also, upon occurrence of Alerts/Events/Emergency, the corresponding data is sent to the cloud.

(Note: The communication frequency is stated for all the streams in “[Default Communication frequency for Telemetry Streams](#)”)

The following is the summary of the data to be sent based on the operating mode of the vehicle

Vehicle State	Stream	Car Type	Communication frequency type
Ignition on and running	Telemetry Streams – Vehicle, Location	ICE	Periodic
	Event/Alert/Emergency		Event based

12. Telemetry Streams

SB API Interface ID : SB-TELEMETRY_STREAMS-001

As Telemetry streams are continuous and periodic, decoding the CAN frames in Dongle and setting the values in proto buffers would increase the size and overhead on the payload sent from Dongle to Cloud.

Hence Vehicle Telemetry streams shall be sent as CAN frames from Dongle and decoded in the CVIP cloud.

Based on communication frequency and sampling frequency, these frames are accumulated and then sent to cloud.

Types of telemetry streams in CVIP

- Vehicle Telemetry – CAN messages are sent in Protobuf directly
- Location Telemetry – decoded messages are sent as Protobuf

Sample Payload for Vehicle Telemetry

Note: This is the semantic JSON representation of the data. Actual data is binary encoded based on Protobuf schema.

Identifier field shall be communicated in Decimal format. Dongle shall read the CAN ID in Hex format and perform Hex to Decimal conversion and communicate

message_id: "dd52bab1-c64a-431b-a945-2aa14be0eb03"

e_Tbox_application_state: provisioned

Tbox_esim_state: normal_sim

version: "2.0.0"

time_stamp {

seconds: 1624946458

nanos: 268381000

}

canDataPayload {

canData {

canDataPacket {

identifier: 296

```

    dlc: 8
    cData {
      pdu: "\000\000\000\002\002\200\000\000"
      timestamp {
        seconds: 1624946458
        nanos: 267869000
      }
    }
    cData {
      pdu: "\000\000\000\002\002\200\000\000"
      timestamp {
        seconds: 1624946458
        nanos: 267963000
      }
    }
    cData {
      pdu: "\000\000\000\002\002\200\000\000"
      timestamp {
        seconds: 1624946458
        nanos: 268031000
      }
    }
  }
}

```

Default Communication frequency for Telemetry Streams

12.1.1 Communication Frequency:

Communication frequency: The frequency at which message is transmitted to cloud

12.1.2 Sampling Frequency:

Sampling frequency: The frequency at which Dongle collates/receives the CAN frames.

Streams	Communication Frequency in seconds (Default)
---------	---

Vehicle Telemetry	60
Location Telemetry	5

12.2 Vehicle Telemetry

SB API Interface ID : SB-TELEMETRY_STREAMS-002

Default Communication frequency: 60 seconds

Topic: /dongle/{vinnum}/{protocol}{dataformat}/vehicleTelemetry

List of frames under Vehicle Telemetry (ICE):

Frame ID	Frame Name	DLC	Sampling frequency in ms (Default)
0x356	STATUS_BH_BCM1	8	250ms
0x46C	STATUS_BH_BCM2	8	500ms
0x3E2	STATUS_CCAN3	8	100ms
0x760	TRIP_A_B	6	1000ms
0x3E6	STATUS_CCAN5	8	100ms
0x46C	STATUS_BH_BCM2	8	500ms
0x46C	STATUS_BH_BCM2	8	500ms

12.3 Location Telemetry

SB API Interface ID : SB-TELEMETRY_STREAMS-003

Default Communication frequency: 5 seconds

Topic: /dongle/{vinnum}/{protocol}{dataformat}/locationTelemetry

Sample Payload for Location Telemetry

Note: This is the semantic JSON representation of the data. Actual data is binary encoded based on Protobuf schema.

message_id: "dd52bab1-c64a-431b-a945-73b75119ab42"

e_Tbox_application_state: provisioned

Tbox_esim_state: normal_sim

version: "2.0.0"

signalStrength

locationPayload {

locationData {

time_stamp {

seconds: 1624946278

nanos: 258217000

```

}
gpsLat: 146.37778917295014
gpsLong: -103.84493212237936
gpsCourseAngle: 337.1260520799143
gpsSignalQuality: 97.36062286424145
gpsFixedStatus:1
speed:25
}

```

13. Vehicle Alerts & Events

SB API Interface ID : SB-ALERTS_EVENTS-001

Alerts and Events are to be processed real-time; hence it has to be sent as decoded value (not as CAN frames) to CVIP.

The following are the list of items considered under Alerts & Events in CVIP

- Vehicle Event
- Vehicle Alerts
- Emergency & Safety Alerts
- Diagnostic Alerts
- **Driving Score**

13.1 Vehicle Alerts

SB API Interface ID : SB-ALERTS_EVENTS-002

Alert is an occurrence of a particular event that is sent to CVIP for action or notification. Alert payloads must be self-sufficient to trigger appropriate actions or notifications.

alertID shall be generated using the same logic used for message ID generation.

Refer Section 4: Glossary

Sample Payload

```

message_id: "dd52bab1-c64a-431b-a945-24750675772e"
e_Tbox_application_state: provisioned
Tbox_esim_state: normal_sim
version: "2.0.0"
time_stamp {
  seconds: 1635577153
}

```

```
    nanos: 493990000
  }
  alertPayload {
    AlertsData {
      time_stamp {
        seconds: 1635577153
        nanos: 494066000
      }
      alertState: alsAlert
      alertType: SpeedAlert
      alertID: "cba73860-1555-11ec-8ed9-c94385ba7cbe"
      isLive: true
    }
    SpeedAlert {
      vehicleSpeed: 90.0
      gpsPayload {
        gpsLat: 12.971598
        gpsLong: 77.594566
        gpsAlt: 920.5
        gpsAccLat: 1.2
        gpsAccLon: 1.3
        gpsAccAlt: 2.0
        gpsCourseAngle: 180.0
        gpsSignalQuality: 4.5
        gpsFixedStatus: 1
      }
    }
  }
}
```

NOTE : For Live alerts if any of the conditions mentioned in the calibration parameters and Algorithm for alert processing column is not satisfied, please set isLive = False for communication to CVIP.

For Live alerts if all the conditions mentioned in the calibration parameters and Algorithm for alert processing column is satisfied, please set isLive = true for communication to CVIP.

Each Live Alert instance is to be uniquely identified by an AlertId. For any specific Alert if isLive is set to true, then it has to be mandatorily set to false when the conditions are not met. This will ensure that the mobile app is displaying the correct status of the alert.

13.1.1 Variables used in alert payload

13.2 Vehicle Events

SB API Interface ID : SB-ALERTS_EVENTS-003

An event can be defined as the change in state of behavior of a parameter which needs to be sent

Sno	Alert	Description
0	InvalidAlertType	Invalid alert
3	SpeedAlert	Alert if vehicle crosses a pre-defined speed
10	TowAwayAlert	Alert if vehicle is towed away (no ignition, parking brake applied, but vehicle moves)
14	Engine Idling Alert	Alert when Engine is Idling
16	Parking Disturbance Alert	Alert if vehicle is moved while in parking condition
17	DongleStatusAlert	Alert when the dongle status is disconnected
19	Harsh Acceleration Alert (Part of Trip Information)	Alert if vehicle is driven in rash manner
20	Harsh Brake Alert (Part of Trip Information)	Alert if vehicle is driven in rash manner
21	Harsh Turn Alert (Part of Trip Information)	Alert if vehicle is driven in rash manner
28	RemoteImmobilizationAlert	Alert when the Remote immobilization occurs

to CVIP for reporting the current state. The event payload shall be communicated every time the corresponding CAN signal changes state.

Sample Payload

message_id: "dd52bab1-c64a-431b-a945-25ba912c4a26"

e_Tbox_application_state: provisioned

Tbox_esim_state: normal_sim

version: "2.0.0"

eventPayload {

EventsData {

```

time_stamp {
    seconds: 1624946270
    nanos: 463270000
}
TboxStateUpdateEvent {
    previousState: Provisioned
    CurrentState: Customer
}
}

```

13.2.1 Variables used in event payload

1. isRemoteEvent

- isRemoteEvent of type Boolean
- isRemoteEvent has to be true If action has be taken because of remote commad
- isRemoteEvent has to be false If action has be taken because of vehicle key

Event Id	Event Name	Enum associated	Description	IGN ON/OFF
1	IgnitionStatusEvent	[Initialization, IGN_LK, ACC, RUN, START, SNA]	Ignition status of the vehicle	IGN ON & Off
2	MainWakeupEvent	[invalid RCD, No_main_wakeup_req, main_wakeup_req, Wake_up_not_valid]	Main Wakeup request status	IGN ON
5	RemoteBlinkerEvent	defaultBlinkersStatus=0; invalidBlinkersStatus =1; TurnBlinkers_on=2; TurnBlinkers_off =3;	Remote Blinker Status	Remote Blinker Status on and off isRemoteEvent of type boolean
6	Remote Door	invalidRemoteDoorLockUnlockCommand=0;	Notification on remote	All doors should be locked and

Event Id	Event Name	Enum associated	Description	IGN ON/OFF
	LockUnlockCommand Event	No_Request=1; Request_Successful_All_Doors_Locked=2; Request_Successful_All_Door_Unlocked=3;	door lock unlock command	unlocked

13.3 Diagnostic Alerts

SB API Interface ID : SB-ALERTS_EVENTS-004

All Diagnostic alerts are expected to be sent by Dongle to CVIP if any of the alert is active for each ignition off to on transition. The alert shall also be communicated if the alert status changes to off to on state.

S.No.	Alert	Description	Actual value	CAN Signal Condition Check	Debounce Time
0	InvalidDiagAlert	Invalid Diagnostics Alert	NA	NA	NA
1	MIL lamp ON warning	Alert when MIL lamp lightning fixed	0 : No MIL lamp lightning fixed request 1 : MIL lamp lightning fixed request	Continuous on	Immediate

13.4 Emergency & Safety Alerts

SB API Interface ID : SB-ALERTS_EVENTS-005

AlertID	Emergency Alert	Description
0	Invalid Emergency Alert	Invalid Emergency Alert
1	Auto Crash Notification Emergency Alert	Crash info detected

Sample Payload

```
EmergencyAlertData {
  time_stamp {
```

```

    seconds: 1623947052
    nanos: 553492000
  }
  emergencyalertType: CrashInfo
  CrashInfo {
    gpsLat: 5555.0
    gpsLong: 1234.0
    gpsAlt: 1241.0
  }
}
EmergencyAlertData {
  time_stamp {
    seconds: 1623947052
    nanos: 553524000
  }
  emergencyalertType: CrashInfo
  CrashInfo {
    gpsLat: 5255.0
    gpsLong: 4255.0
    gpsAlt: 2000.0
  }
}

```

13.5 Alert Classification

SB API Interface ID : SB-ALERTS_EVENTS-006

NOTE: The below alert classification table will be determined after discussion with **DONGLE MANUFACTURER** and will be added in next version of the SB interface document.

S.No	Alerts	Category
15	MILLampOnAlert	Diagnostic
31	SpeedAlert	General
39	ParkingDisturbanceAlert	critical
40	TowAwayAlert	critical
43	DongleStatusAlert	critical
48	HarshAccelerationAlert	General
49	HarshBrakingAlert	General
50	SharpTurnAlert	General

13.6 Driving

score

SB API Interface ID : SB-DRIVING_SCORE-007

The driving score shall be computed within the Dongle and sent to CVIP aspart of Trip End Payload. CVIP aggregates the driving score per week for each vehicle and send to the application end to display.

13.7 Trip Parameter Calculation

SB API Interface ID : SB-TRIP_TELEMETRY-001
Dongle is expected to calculate listed below parameters for a trip and communicate three different types of payloads to CVIP.

- Trip Start Payload – Payload generated and communicated at the start of the trip
- Trip Progress Payload – Payload generated & communicated at a periodicity of 60sec during ignition on
- Trip End Payload – Payload generated at the creation of new trip and provides details of the previously completed trip parameters

13.8 MQTT Topics

SB API Interface ID : SB-TRIP_TELEMETRY-002

Payload Name	MQTT Topics
Trip Start	/dongle/{vinnum}/{protocol}{dataformat}/tripStart
Trip Progress	/dongle/{vinnum}/{protocol}{dataformat}/tripCurrent
Trip End	/dongle/{vinnum}/{protocol}{dataformat}/tripEnd

13.9 Trip Parameters

SB API Interface ID : SB-TRIP_TELEMETRY-003

Parameter	Description	Condition for calculation	Datatype	Mandatory/Optional	Short key Name
Trip Id	Unique id for the trip	Shall be configured as DDMMYYYYNo. For Eg. If the current date is 26-Nov-20, and the trip no is 01, then the trip id shall be configured as 2611202001. For a date, the trip no. shall start with 01 and get incremented for new trips like 2611202002 as second trip, 2611202003 for third trip. If the date changes, then the trip no shall be be start again from 01. All trips shall be uniquely identified by trip id.	Integer	Mandatory	tripId

Start Date	Start date and time of the trip	start date and time in epoch format. This is the time at which the ignition was switched on and the time difference between the previous ignition off event and ignition on event is more than 120 mins. If the time difference is less than 120 mins, then the start date shall continue with previous trips start date. This is the time at which the ignition was switched on and the time difference between last ignition off event and current ignition on event is more than 120 mins.	Integer	Mandatory	startTime
End Date	End date and time of the trip	end date and time in epoch format. This is the time at which the ignition was switched off.	Integer	Mandatory	endTime
Current Date	Current date and time of the trip when the payload is communicated.	Current date and time in epoch format. This is the time at which the tripCurrent Payload is communicated from Dongle to CVIP	Integer	Mandatory	currentTime
End Location	End location of the trip	End location information in GPS coordinates. For Eg. gnssInfoEnd { "gpsLat": "13.084534", "gpsLong": "18.063455", "gpsAlt": "2.0245671", "gpsAccLat": "0.001", "gpsAccLong": "0.001", "gpsAccAlt": "0.1", "gpsCourseAngle": "25", "gpsSignalQuality": "1.1", "gpsFixedstatus" = 0 or 1 }	Json Object	Mandatory	gnssInfoEnd
Current Location	Current Location of the trip	Current location information in GPS coordinates. For Eg. gnssInfoCurrent { "gpsLat": "13.084534", "gpsLong": "18.063455", "gpsAlt": "2.0245671", "gpsAccLat": "0.001", "gpsAccLong": "0.001", "gpsAccAlt": "0.1", "gpsCourseAngle": "25", "gpsSignalQuality": "1.1", "gpsFixedstatus" = 0 or 1 }	Json Object	Mandatory	gnssInfoCurrent

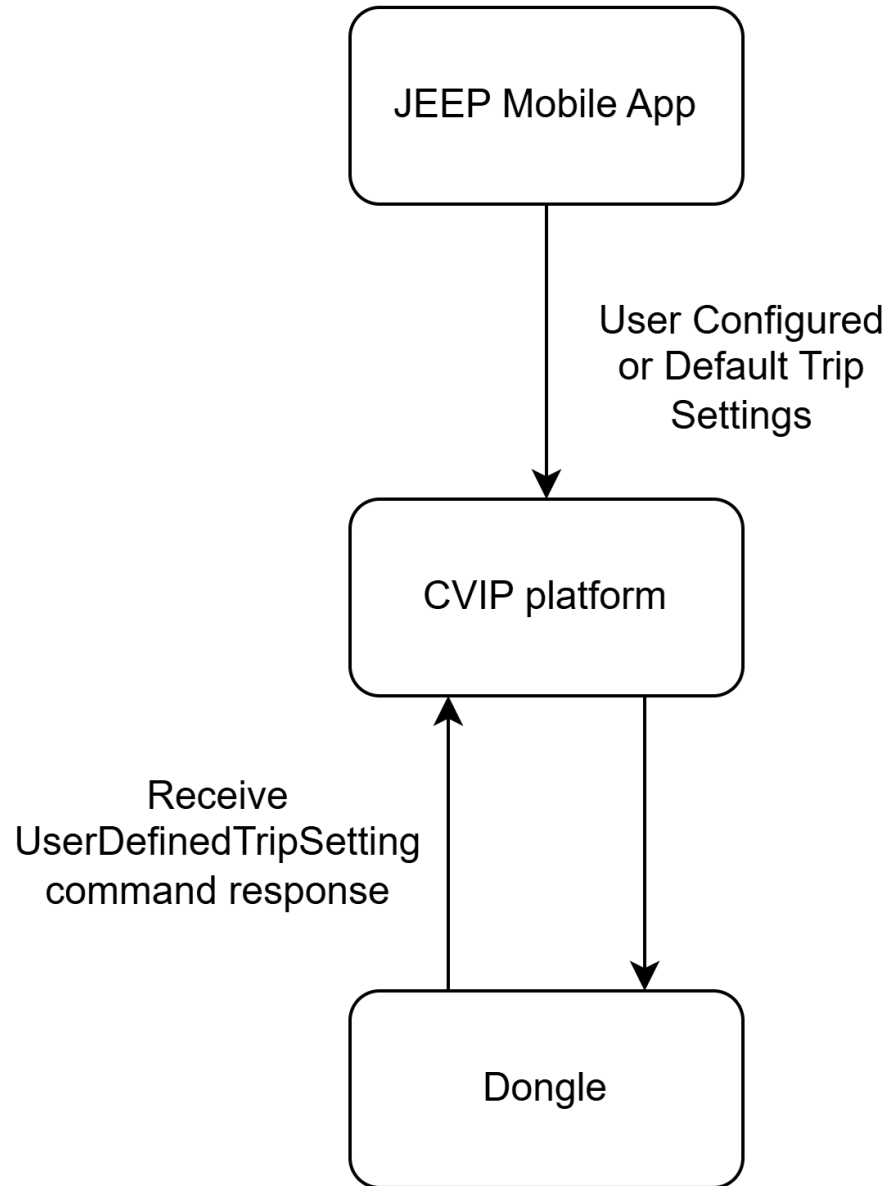
Start Location	Start Location of the trip	Start location information in GPS coordinates. For Eg. gnssInfoStart { "gpsLat": "13.084534", "gpsLong": "18.063455", "gpsAlt": "2.0245671", "gpsAccLat": "0.001", "gpsAccLong": "0.001", "gpsAccAlt": "0.1", "gpsCourseAngle": "25", "gpsSignalQuality": "1.1", "gpsFixedstatus" = 0 or 1 }	Json Object	Mandatory	gnssInfoStart
Distance travelled	Total distance travelled till current location	Distance travelled in a trip computed in km using the signal km_total as received in Frame id 0xF6.	Float	Mandatory	tripDistance
Time Travelled	Trip start time to current time	Difference between the trip start time and trip end time in mins.	Integer	Mandatory	tripTime
Trip Type	[Active / Idle] if one value of speed >=5km/hr it is an Active trip	If vehicle speed (DRV_CLUSTER_DSPEED) < 5kmph through the trip, then it is Idle. If vehicle speed exceeds 5kmph for more than 200ms then it is Active	String	Mandatory	tripType
Hard Brake	Number of hard brakes applied	No of events when the Vehicle speed Gradient > hardBrakeThr (Configurable DID). Vehicle speed gradient is the difference between vehicle speed logged before 200ms and current vehicle speed. Each event of hard braking shall be separated by a time duration of 1 sec.	Integer	Mandatory	hardBrakeCnt
Harsh Acceleration	Number of hard accelerations applied	No of events when the Vehicle speed Gradient > hardAccThr (Configurable DID). Vehicle speed gradient is the difference between Current vehicle speed and vehicle speed logged before 200ms. Each event of hard acceleration shall be separated by a time duration of 3 sec.	Integer	Mandatory	harshAccCnt
High Speed	Number of times in high speed	No of events when the vehicle speed (DRV_CLUSTER_DSPEED) has exceeded speed greater than 80kmph and 120kmph in both ICE and EV.	Integer	Mandatory	highSpeedCnt
Harsh Turn	Number of times the harsh turn event	if HarshTurn=1, then HarshTurnAlert to be generated Condition: G1-G2/t > c &	Integer	Mandatory	harshTurnCnt

	occurred	EFCMNT_PDLE_ACCEL > 30% then HarshTurn=1. (G1, G2 in deg/s) & Vhl Speed > p km/hr			
Average Speed	Average speed for a particular trip	Trip specific average speed	Float	Mandatory	speedAvg
Top Speed	Top Speed for a particular trip	Trip specific top speed	Integer	Mandatory	topSpeed

13.10 Logic/Flowchart for Trip Parameter Configuration

SB API Interface ID : SB-TRIP_TELEMETRY-004

13.10.1 Trip Settings Configuration Logic



13.10.2 Trip Creation/Trip ID Generation Logic

The Trip Creation/ Trip ID Generation Logic is computed in the dongle and sent to CVIP in Trip Telemetry. The User Defined Trip Settings are sent to dongle from CVIP as a command.

13.10.3 Assumptions

- The trip start shall be send by the Dongle only when the predefined conditions for trip like minimum trip distance and idle time is valid. Default value shall be considered if no settings are received from CVIP or App.

13.11 Trip Start Payload:

SB API Interface ID : SB-TRIP_TELEMETRY-005

This payload shall be communicated by Dongle at the start of the trip. If the payload is not communicated at start of the trip, then the Dongle shall buffer this payload and communicate once the network is latched.

Parameter	Description	Short Name
Trip Id	Unique id for the trip	tripId
Start Date	Start date and time of the trip	startDate
Start Location	Start location of the trip	gnssInfoStart

Note: This is the semantic JSON representation of the data. Actual data is binary encoded based on Protobuf schema.

message_id: "dd52bab1-c64a-431b-a945-faa3adc29c06"

e_Tbox_application_state: provisioned

Tbox_esim_state: normal_sim

version: "2.0.0"

time_stamp {

seconds: 1627985481

nanos: 882646000

}

tripPayload {

tripstartpayload {

tripstartdata {

tripId: 3007202101

start_time {

seconds: 1627985481

```
    nanos: 882684000
  }
  start_date: 30072021
  gnssinfostart {
    startlocationData {
      gpsLat: 100.0
      gpsLong: -103.84493212237936
      gpsAlt: 2239.0
      gpsAccLat: 439.0
      gpsAccLon: 840.0
      gpsAccAlt: 589.0
      gpsCourseAngle: 337.1260520799143
      gpsSignalQuality: 97.36062286424145
      gpsFixedStatus: 1
    }
  }
}
```

13.12 Trip Progress payload

SB API Interface ID : SB-TRIP_TELEMETRY-006

This payload shall be communicated at a periodicity of 60s. If the Dongle is unable to communicate due to network failure, then the payload shall be buffered and communicated once the network resumes. The timestamp of buffered payload shall hold the same time at which the payload was generated and shall not be the time at which payload was communicated.

Parameter	Description	Short Name
Trip Id	Unique id for the trip	tripId
Current Date	Start date and time of the trip	currentTime
Current Location	Start location of the trip	gnssInfoCurrent
Distance travelled	Total distance travelled till current location	tripDistance
Time Travelled	Trip start time to current time	tripTime
Trip Type	[Active/Idle] if one value of speed ≥ 5 km/hr it is an Active trip	tripType
Hard Brake	Number of hard brakes applied	hardBrakeCnt
Harsh Acceleration	Number of hard accelerations applied	harshAccCnt
High Speed	Number of times in high speed. High speed should be considered for both speed greater than 80kmph and 120kmph in both ICE and EV.	highSpeedCnt
Harsh Turn	Number of times harsh turn events occurred	harshTurnCnt
Average Speed	Trip specific average speed	speedAvg
Top Speed	Trip specific top speed	topSpeed

Note: This is the semantic JSON representation of the data. Actual data is binary encoded based on Protobuf schema.

message_id: "dd52bab1-c64a-431b-a945-a546528f68df"

Tbox_operating_state: normal

e_Tbox_application_state: provisioned

Tbox_esim_state: normal_sim

version: "2.0.0"

time_stamp {

seconds: 1627985798

nanos: 507355000

}

tripPayload {

tripinprogresspayload {

tripinprogressdata {

tripId: 3007202101

```
current_time {
  seconds: 1627985798
  nanos: 507407000
}
current_date: 30072021
gnssinfo {
  currentlocationData {
    gpsLat: 100.0
    gpsLong: -103.84493212237936
    gpsAlt: 2239.0
    gpsAccLat: 439.0
    gpsAccLon: 840.0
    gpsAccAlt: 589.0
    gpsCourseAngle: 337.1260520799143
    gpsSignalQuality: 97.36062286424145
    gpsFixedStatus: 1
  }
}
tripDistance: 56.0
tripTime: 45
tripType: "Active"
hardBrakeCnt: 2
harshTurnCnt: 4
avgSpeed: 75.0
topSpeed: 86.0
}
tripinprogressdata {
  tripId: 3007202102
  current_time {
    seconds: 1627985798
    nanos: 507432000
  }
  current_date: 30072021
  gnssinfo {
    currentlocationData {
      gpsLat: 100.0
      gpsLong: -103.84493212237936
      gpsAlt: 2239.0
```

```

    gpsAccLat: 439.0
    gpsAccLon: 840.0
    gpsAccAlt: 589.0
    gpsCourseAngle: 337.1260520799143
    gpsSignalQuality: 97.36062286424145
    gpsFixedStatus: 1
  }
}
tripDistance: 56.0
tripTime: 45
tripType: "Active"
harshAccCnt: 1
highSpeedCnt: 2
harshTurnCnt: 4
avgSpeed: 70.0
topSpeed: 80.0
}
}
}

```

13.13 Trip End Payload

SB API Interface ID : SB-TRIP_TELEMETRY-007

This payload shall be communicated when a new trip is created due to duration between ignition off to on greater than 120 mins. The trip end payload shall communicate the parameters previously completed payload. If the payload is not able to get communicated due to network failures, then the Dongle shall buffer the payload and communicate once the network resumes.

Parameter	Description	Short Name
Trip Id	Unique id for the trip	tripId
End Date	End date and time of the trip	endTime
End Location	End location of the trip	gnssInfoEnd
Average energy Efficiency	Fuel Consumption – Trip specific fuel economy (ICE)	avgFuelEconomy (ICE)
Distance travelled	Total distance travelled till current location	tripDistance
Time Travelled	Trip start time to current time	tripTime
Hard Brake	Number of hard brakes applied	hardBrakeCnt
Harsh Acceleration	Number of hard accelerations applied	harshAccCnt

High Speed	Number of times in high speed. High speed should be considered for both speed greater than 80kmph and 120kmph	highSpeedCnt
Harsh Turn	Number of times harsh turn events occurred	harshTurnCnt
Average Speed	Trip specific average speed	speedAvg
Top Speed	Trip specific top speed	topSpeed

Note: This is the semantic JSON representation of the data. Actual data is binary encoded based on Protobuf schema.

message_id: "dd52bab1-c64a-431b-a945-a546528f68df"

Tbox_operating_state: normal

e_Tbox_application_state: provisioned

Tbox_esim_state: normal_sim

version: "2.0.0"

time_stamp {

seconds: 1627985798

nanos: 507355000

}

tripPayload {

tripendedpayload {

tripenddata {

tripId: 3007202101

end_time {

seconds: 1627985798

nanos: 507407000

}

end_date: 30072021

gnssinfoend {

endlocationData {

gpsLat: 100.0

gpsLong: -103.84493212237936

gpsAlt: 2239.0

gpsAccLat: 439.0

gpsAccLon: 840.0

gpsAccAlt: 589.0

gpsCourseAngle: 337.1260520799143

gpsSignalQuality: 97.36062286424145

gpsFixedStatus: 1

}

```

    }
    tripDistance: 56.0
    drivingScore : 80
    tripTime: 45
    idleDuration: 20
    idlingCnt: 5
    hardBrakeCnt: 2
    harshTurnCnt: 4
    avgSpeed: 75.0
    topSpeed: 86.0
  }
}

```

13.14 Trip Payload Prioritization:

SB API Interface ID : SB-TRIP_TELEMETRY-008

For a unique **Trip_ID**, Dongle should first send the **Trip start payload**, which should be followed by **Trip in progress payloads** and **Trip End payload**.

1. Trip start
2. Trip in progress
3. Trip end

In case of network failure Dongle should send Buffered data.

This Buffered data should have the details in the following priority

➔ Current trip details

1.Trip Start

2.Trip In-progress

➔ Previous Trips completed

1.Previous Trip Start

2. Previous Trip In-progress

3. Previous Trip End

...

1. Previous Trip Start

2. Previous Trip In-progress

3. Previous Trip End

14. Annexure

14.1 CAN mapping:

Message_Name	Signal_Name	ECU_Name
STATUS_BH_BCM1	FuelLevel	BCM
STATUS_BH_BCM2	BatteryVoltageLevel	BCM
STATUS_CCAN3	EngineWaterTemp	BCM
TRIP_A_B	TotalOdometer	BCM
STATUS_CCAN5	EngineSpeed	BCM
STATUS_BH_BCM2	ExternalTemperatureC	BCM
STATUS_BH_BCM2	ExternalTemperatureF	BCM

14.2 CAN Signal Mapping Reference

Refer CANFrames_jeeplife_v0.1

15. References

- [1] FCA_JEEPLIFE_User Manual3.1_22-June-2021
- [2] HarmanAPISpecificationV1.1
- [3] M6 Dongle CAN mapping sheet v1.1
- [4] Stellantis_Neoway Dongle_Feature List_V1